

MAURER–CARTAN ELEMENTS IN THE PILLOWCASE AND THE BAR-NATAN INVARIANT FOR THE $(-2, 3, q)$ PRETZEL KNOTS

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ABSTRACT. Cazassus, Herald, Kirk and Kotelskiy conjectured that the reduced singular instanton homology of a knot is recovered from any decomposition along a Conway sphere by Lagrangian Floer homology in the pillowcase, once the immersed curves of the two tangles carry suitable bounding cochains. For Smith’s decomposition of the pretzel knots $P(-2, 3, q)$, which separates a rational tangle carrying the earring from the sum of two other rational tangles, zero bounding cochains give the wrong rank. Working in the algebraic model of Kotelskiy, Watson and Zibrowius, where the Maurer–Cartan equation is a finite identity, we construct for $P(-2, 3, 5)$ and $P(-2, 3, 7)$ explicit Maurer–Cartan elements, each obtained by smoothing a model of Smith’s curve at a single self-intersection, that restore the rank. For $P(-2, 3, 7)$, a second closure of the tangle selects one of them, conditionally on the conjecture and on a hypothesis about the support of the cochain. The resulting object is a rational arc together with a closed curve carrying a two-dimensional local system, and one further term near a corner turns it into the Bar-Natan invariant of the tangle. Granting one case of an intersection formula of Kotelskiy, Watson and Zibrowius, we show that no rational closure distinguishes the two. We ask whether the object that the conjecture assigns to this tangle is its Bar-Natan invariant, and record evidence on both sides.

1. INTRODUCTION

1.1. Bounding cochains in the pillowcase. Let $K \subset S^3$ be a knot. On the gauge-theoretic side sits the reduced singular instanton knot homology $I^\natural(K)$ of Kronheimer and Mrowka [KM11a, KM11b]. On the symplectic side sits the pillowcase program of Hedden, Herald and Kirk [HHK14, HHK18]. Cut K along a Conway sphere into two 2-tangles. The traceless $SU(2)$ character variety of the four-punctured sphere is the *pillowcase*, a two-sphere with four orbifold points, and after a holonomy perturbation each tangle determines an immersed curve in it; the tangle that carries the earring of the reduced theory gives a figure-eight curve. Hedden, Herald and Kirk conjecture that for a suitable decomposition the Lagrangian Floer homology of the two curves is isomorphic to $I^\natural(K; \mathbb{F}_2)$ [HHK18, Conj. 6.5]. For general decompositions figure-eight bubbles at the self-intersections of the curves deform the differential. Cazassus, Herald, Kirk and Kotelskiy conjecture that each tangle is assigned an immersed curve together with a bounding cochain, and that the deformed theory recovers $I^\natural(K)$ [CHKK22, Conj. D]. Conditional on this conjecture, Smith proves that the cochain on one Conway-sum tangle in $P(-2, 3, 5)$ cannot vanish [Smi24, proof of Thm. 5.9].

This paper studies Smith’s decomposition of the pretzel knots $K_q = P(-2, 3, q)$, $q \geq 3$ odd (2.3), as the numerator closure of a sum of three rational tangles, $K_q = \text{num}(Q_{-1/2} + Q_{1/3} + Q_{1/q})$, with the earring on the -2 band. Smith shows that the Floer homology with zero bounding cochain has the wrong rank at $q = 5$ [Smi24, proof of Thm. 5.9], and at $q = 7$ the same holds for a piecewise-linear model of his curve (Proposition 3.7). We work over the algebra $\mathcal{B}$

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of Kotelskiy, Watson and Zibrowius [KWZ19], the endomorphism algebra of two arcs in the four-punctured sphere. For $q = 5, 7$ we construct Maurer–Cartan elements for the twisted complex of the model over $\mathcal{B}$ that correct the rank, and we identify the resulting objects with the Bar-Natan invariant of the tangle $Q_{1/3} + Q_{1/q}$ up to one further term (Theorems 1.1 and 1.2). The companion paper [Wue26] computes $I^\natural(K_q; \mathbb{Z}) \cong \mathbb{Z}^{q+2}$ for every odd $q \geq 3$ [Wue26, Thm. 1.1]; these instanton dimensions are the targets of the pairing dimensions computed here. It also shows that for a different decomposition of K_q , obtained from the tangle of Hedden, Herald and Kirk for $T(3, 5)$ by regluing, the pillowcase Floer homology with zero bounding cochain is isomorphic to $I^\natural(K_q; \mathbb{F}_2)$ as a $\mathbb{Z}/4$ -graded vector space, for odd $q \geq 7$ [Wue26, Thm. 1.2].

1.2. Smith’s decomposition. In Smith’s decomposition the earring is placed on the summand $Q_{-1/2}$, which gives the tangle $\widehat{Q_{-1/2}}$; for a slope r we write $\widehat{Q_r}$ for the tangle Q_r with an earring, whose curve E_r surrounds the line of Q_{-r} (§§3.1 and 4.1). The other tangle is $T_q = Q_{1/3} + Q_{1/q}$. We work in Smith’s coordinates on the pillowcase, with his holonomy perturbation for small negative values of its parameter t (§2.2). Following Cazassus, Herald, Kirk and Kotelskiy [CHKK22, §11], we encode curves as twisted complexes over $\mathcal{B}$. The higher products of $\mathcal{B}$ vanish, so the Maurer–Cartan equation for a deformation b of a twisted complex (V, δ) is the finite identity $(\delta + b)^2 = 0$. The pairing dimension of two twisted complexes is the dimension of the homology of their morphism complex, a Floer homology dimension for curves in the position required by [KWZ19, Thm. 5.22] (§2.3). We write N_q for the twisted complex of a piecewise-linear model of Smith’s curve for T_q , and $\mathcal{E}_r$ for that of the earring curve E_r , with $\mathcal{E} = \mathcal{E}_{-1/2}$.

Under the equivalence of $\text{Tw}(\mathcal{B})$ with the twisted complexes over the relevant wrapped Fukaya category of the pillowcase [CHKK22, §11.5], the deformed complexes (N_q, b) play the role of immersed curves with bounding cochains. The switch of a smoothing at a self-intersection, defined below, is the twisted-complex form of the Lagrangian surgery there, which in immersed Floer theory corresponds to a degree-one bounding cochain at that point (for curves on surfaces compare Auroux and Smith, §5.1). We do not express the switches in the immersed Floer complex of Akaho and Joyce [AJ10] (Remark 3.16).

For $q = 7$ the pairing dimension of $\mathcal{E}$ with N_7 is 7, not $9 = \dim I^\natural(K_7)$ (Proposition 3.7). Each self-intersection of the model lifts to two double points of its lift to the torus double cover, exchanged by the elliptic involution; we call the pair a *self-intersection orbit*. A smoothing of the model at a self-intersection orbit that keeps the generators of N_7 changes its differential by a matrix b_s , the *switch* of the smoothing. At four self-intersection orbits, labelled $S_{18}, S_{25}, S_{69}, S_{74}$ in §3.3.3, smoothing gives a switch b_s that is a Maurer–Cartan element and raises the pairing dimension to 9, and the four objects (N_7, b_s) fall into three homotopy-equivalence classes (Corollary 3.9). Replacing $\widehat{Q_{-1/2}}$ by $\widehat{Q_{-3/4}}$ closes T_7 into a second knot K_* . Its reduced Khovanov homology is free of rank 31, which equals the coefficient norm of its Alexander polynomial, so the bounds of Kronheimer and Mrowka [KM10, KM11b] give $I^\natural(K_*; \mathbb{Z}) \cong \mathbb{Z}^{31}$, and the pairing dimensions with the second earring separate the three classes (Proposition 3.8).

Conditional selection at $q = 7$. Assume the conjecture of Cazassus, Herald, Kirk and Kotelskiy, and assume Hypothesis 3.13: the assigned object is homotopy equivalent to the oriented smoothing of the model at a single self-intersection orbit, or to the deformation of N_7 by a combination of the four switches above. Then the object assigned to T_7 is homotopy equivalent to the common class of $(N_7, b_{S_{18}})$ and $(N_7, b_{S_{25}})$ (Corollary 3.14). The orbits S_{18} and S_{25} are the two at which a double point cuts the lift into two loops the shorter of which has torus class $(2, 2)$ up to sign (Proposition 3.10). For positive t two of the four smoothings give 5 and 7 instead of 9, and the two-closure selection holds for both signs (Remark 3.17). We also show that Gao’s wrapped representability theorem [Gao17] cannot be applied to the Lagrangian correspondence induced by the line of $Q_{1/3}$, which is immersed with a triple point (Proposition 3.4).

Kotelskiy, Watson and Zibrowius assign to T_q its Bar-Natan invariant $\widetilde{\text{BN}}(T_q)$, an immersed multicurve with local systems in the four-punctured sphere [KWZ19]. Drawn in Smith's coordinates it becomes an object BN_q of $\text{Tw}(\mathcal{B})$ (§4.1). For $r \in \mathbb{Q} \cup \{\infty\}$ let $K_r(q) = \text{num}(Q_r + T_q)$, so that $K_{-1/2}(q) = K_q$. A compact curve in the pillowcase is *linear* if its lifts to the plane lie close to parallel straight lines and are graphs over them (Definition 4.4). Two twisted complexes are *strictly isomorphic* if a bijection of their generators that preserves idempotents carries one differential to the other entry by entry (§2.3). For odd q put $r_q = (1 - q)/(3q - 4)$. Let α_q be the straight arc between $(0, 0)$ and $(\pi, 0)$ that the earring curve E_{r_q} surrounds, and let c be the boundary of a regular neighborhood of the straight arc that E_{-1} surrounds; c separates $\{(\pi, 0), (0, \pi)\}$ from $\{(0, 0), (\pi, \pi)\}$ (§4.2). For $q \in \{5, 7\}$ let b be one of the selected elements $b_{S_{18}}, b_{S_{25}}$ at $q = 7$, or at $q = 5$ the switch b_5 of a smoothing that raises the pairing dimension with $\mathcal{E}$ from 5 to 7 (§4.2).

Theorem 1.1. *Let $q \in \{5, 7\}$.*

- (1) *The object (N_q, b) is strictly isomorphic to $\alpha_q \oplus (c, J_2)$, where J_2 is the indecomposable two-dimensional local system on c .*
- (2) *There is a term $\kappa_q \in \text{End}(N_q)$, which replaces one arrow of the differential near the corner $(0, 0)$, such that $b + \kappa_q$ is a Maurer–Cartan element and $(N_q, b + \kappa_q)$ is strictly isomorphic to BN_q .*

For every odd q with $5 \leq q \leq 21$, BN_q is an arc obtained from α_q by sliding one of its ends along c .

Theorem 1.2. *Let $5 \leq q \leq 21$ be odd, and let Γ be a linear curve parallel neither to α_q nor to c . Then the minimal intersection numbers satisfy $i(\Gamma, \text{BN}_q) = i(\Gamma, \alpha_q) + 2i(\Gamma, c)$, and if Γ is bigraded, with the corner (π, π) as the special puncture of [KWZ19], then, granting the arc case of the intersection formula of Kotelskiy, Watson and Zibrowius [KWZ19, Thm. 5.25], $\dim HF(\Gamma, \text{BN}_q) = \dim HF(\Gamma, \alpha_q \oplus (c, J_2))$.*

Corollary 1.3. *Granting the arc case of the intersection formula of Kotelskiy, Watson and Zibrowius [KWZ19, Thm. 5.25], for $q \in \{5, 7\}$ and every $r \in \mathbb{Q} \cup \{\infty\}$,*

$$\dim HF(E_r, (N_q, b)) = \dim HF(E_r, (N_q, b + \kappa_q)) = \text{rank } \widetilde{Kh}(K_r(q); \mathbb{F}_2).$$

Thus, granting that case, no rational closure distinguishes (N_q, b) from BN_q ; curves parallel to c do (Remark 4.8). Section 4 states these results in precise form (Theorems 4.1 and 4.5, Corollary 4.6). The element b is the switch of a smoothing of the model at one self-intersection, a Lagrangian surgery, and already reaches the Khovanov rank, whereas κ_q is supported on chords around the corner $(0, 0)$ and not at a self-intersection of the curve. Kotelskiy, Watson and Zibrowius suggest that the spectral sequences from Khovanov homology to Heegaard Floer and instanton homology might be realized by smoothing self-intersections of $\widetilde{\text{BN}}(T)$ and $\widetilde{Kh}(T)$ [KWZ19, §1.8], and Theorem 1.1 proceeds in the opposite direction, from a smoothing of the curve on the instanton side to the Bar-Natan invariant.

Question 1.4. Let $q \geq 5$ be odd and prime to 3. If the tangle assignment of Cazassus, Herald, Kirk and Kotelskiy is defined for T_q with Smith's perturbation for small $t < 0$, is the object of $\text{Tw}(\mathcal{B})$ that it determines homotopy equivalent to BN_q ?

The evidence for $q = 5, 7$ is Theorem 1.1 and Corollary 1.3. Two observations point the other way. First, BN_7 is not homotopy equivalent to any object allowed by Hypothesis 3.13 (Proposition 4.10). An affirmative answer for $q = 7$ would therefore show that the hypothesis fails, and, granting the arc case of the intersection formula [KWZ19, Thm. 5.25], no rational closure could detect the failure. Second, for $q = 11, 13$ the encoding of a numerical trace of

Smith’s curve has pairing dimension $q + 4$ with $E_{-1/2}$, and for $q = 17, 19$ pairing dimension $q + 8$, against rank $\widetilde{Kh}(K_q) = q + 2$; in finite searches no deformation by a few switches reaches BN_q (Remark 4.11). These searches do not exclude a larger bounding cochain, and we pose the question without conjecturing an answer.

1.3. Outline of the arguments. *The Maurer–Cartan elements at $q = 7$.* For small t , Smith’s perturbed curve contains a component homeomorphic to the fiber product of the lines of $Q_{1/3}$ and $Q_{1/7}$ with each circle that the Conway sum acquires over an edge of the pillowcase replaced by two arcs crossing once, joined crosswise to the rest of the curve (the *cross-resolution* of §2.2), and its other components are circles that can be moved off any fixed complementary curve (Proposition 3.1, after [Smi24]). Performing the same cross-resolution on the fiber product of the two lines, drawn piecewise linearly, gives a curve homeomorphic to that component, the model. Encoding the model by its crossings with the arc system of $\mathcal{B}$ gives, after unit cancellations, the twisted complex $N = N_7$ on 31 generators. Morphism complexes between finite twisted complexes are infinite dimensional, and Lemma 3.5 computes their homology exactly: the morphism complex is a mapping cone onto free modules over the polynomial rings of the three faces, and its homology is finite dimensional exactly when those modules are acyclic over the fraction fields. Each self-intersection of the model has two smoothings; since $\mathcal{B}$ has no higher products, the switch b_s of a smoothing that preserves the generators is a Maurer–Cartan element exactly when $(\delta_N + b_s)^2 = 0$, a finite check. The enumeration of all 82 self-intersection orbits (Propositions 3.10 and 3.11) compares the pairing dimensions with the two earrings against the instanton dimensions 9 and 31.

Theorems 1.1 and 1.2. All objects in the statements are reduced twisted complexes of loop type, and a strict isomorphism between two of them is decided by comparing the words of their components; $\widetilde{BN}(T_q)$ is computed with Zibrowius’s program [Zib]. For Theorem 1.2 we realize each curve by a geodesic in a metric of nonpositive curvature (the Alexander–Berg–Bishop theorem, see [AKP26, Thm. 7.28]), in which minimal intersection numbers are counted locally. A lift of BN_q agrees with the union of a lift of α_q and two periods of a lift of c outside small discs at five lattice points, and a four-case local count at one lattice point (4.2) gives the identity of intersection numbers. For bigraded curves we then use the intersection formulas of Kotelskiy, Watson and Zibrowius [KWZ19, Thm. 5.25] and of Zibrowius [Zib20, Thm. 4.45].

1.4. Scope of claims. Proposition 3.7, Theorems 1.1 and 1.2 and Corollary 1.3 are statements about explicitly presented objects of $\text{Tw}(\mathcal{B})$; they neither assume nor prove Conjecture D of [CHKK22], and they do not identify the object that the conjecture of Cazassus, Herald, Kirk and Kotelskiy assigns to T_q . Proposition 3.1 identifies the homeomorphism type of the main components of Smith’s curve; it is not proved that their twisted complex is N_q . This is corroborated for $q \leq 13$ by numerical traces (Remark 3.15, §A.4), and Hypothesis 3.13 is formulated for the model. Corollary 3.14 is conditional on Conjecture D of [CHKK22] and on Hypothesis 3.13. The second statement of Theorem 1.2, Corollary 1.3 and the second statement of Proposition 4.10 grant the arc case of the intersection formula of Kotelskiy, Watson and Zibrowius [KWZ19, Thm. 5.25], the case of an arc that ends at a puncture, which they assert is treated like the compact case but do not write out (proof of Theorem 4.5); the dimensions of Corollary 1.3 are also computed directly at several hundred slopes. The identifications of N_q and of the switches with the encodings and smoothings of the piecewise-linear curves rest on the floating-point geometry of these curves (Appendix A.4), and so do the second part of Proposition 3.7, Propositions 3.10–3.12, Corollary 3.14 and Proposition 4.10, whereas all statements about the displayed complexes are exact. Statements labelled *Computation* (Appendix A.3) are outputs of finite procedures and are recorded as data.

1.5. Organization. Section 2 recalls the pillowcase, Smith’s coordinates and the algebra $\mathcal{B}$. Section 3 treats Smith’s decomposition with zero bounding cochain and the Maurer–Cartan elements at $q = 7$, and Section 4 relates them to the Bar-Natan invariant. Section 5 discusses related work and open problems, and Appendix A prints the remaining data, records finite polygon counts, and describes the computations.

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2. BACKGROUND

This section fixes notation and collects the results from the literature that are used later. Results of [HHK18] are cited by their numbering in arXiv:1501.00028v1. We use $\mathrm{SU}(2)$ as the group of unit quaternions, so that a traceless element is a purely imaginary unit quaternion. Coefficients are in $\mathbb{F}_2$ unless stated otherwise.

2.1. The pillowcase. Let $G \cong \mathbb{Z}^2 \rtimes \mathbb{Z}/2$ be the group of isometries of $\mathbb{R}^2$ generated by $(\gamma, \theta) \mapsto (\gamma + 2\pi, \theta)$, $(\gamma, \theta) \mapsto (\gamma, \theta + 2\pi)$ and $(\gamma, \theta) \mapsto (-\gamma, -\theta)$. The *pillowcase* is the quotient $P = \mathbb{R}^2/G$, a topological 2-sphere. The quotient map $\mathbb{R}^2 \rightarrow P$ is a branched covering, branched over the images of the lattice points $(\pi\mathbb{Z})^2$, which are the four *corners* of P [HHK18, §3.1]. The rectangle $[0, \pi] \times [0, 2\pi]$ is a fundamental domain; we call $[0, \pi] \times [0, \pi]$ the front face, $[0, \pi] \times [\pi, 2\pi]$ the back face, and the images of the lines $\gamma \in \pi\mathbb{Z}$ the left and right edges. Put $P^* = (\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2)/G$, oriented by $d\gamma \wedge d\theta$. The restriction $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2 \rightarrow P^*$ is a regular covering with deck group G , so paths, loops and immersed discs in P^* lift to $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$; we work with such lifts throughout. The double cover $\mathbb{R}^2/(2\pi\mathbb{Z})^2$ of P is a torus, and P is its quotient by the elliptic involution $\rho(\gamma, \theta) = (-\gamma, -\theta)$.

The pillowcase is the traceless character variety of the four-punctured sphere: the space of conjugacy classes of $\mathrm{SU}(2)$ representations of $\pi_1(S^2 \setminus \{a, b, c, d\})$ sending each of a, b, c, d to a traceless element is homeomorphic to P , and the corners correspond to the abelian representations [HHK14, Prop. 3.1], [HHK18, Prop. 6.1]. We use the coordinates of Smith (§2.2) throughout.

Let $K \subset S^3$ be a knot, cut along a Conway sphere into two 2-tangles. Restricting the traceless representations of each tangle to the four-punctured sphere, after a holonomy perturbation, gives an immersed curve in the pillowcase for each tangle, its perturbed traceless character variety, which may have several components. One tangle carries the *earring*, the small meridional circle that Kronheimer and Mrowka add to the knot to define the reduced theory [KM11b], and its curve is a figure-eight curve [HHK18, HK24]. Hedden, Herald and Kirk define the Lagrangian Floer homology over $\mathbb{F}_2$ of the two curves and conjecture that for a suitable decomposition it is isomorphic to the reduced singular instanton homology $I^\natural(K; \mathbb{F}_2)$ [HHK18, Conj. 6.5]. For an arbitrary decomposition the Floer differential is deformed by figure-eight bubbles at the self-intersections of the curves, and Cazassus, Herald, Kirk and Kotelskiy conjecture that each tangle, including the one that carries the earring, is assigned its perturbed character variety together with a bounding cochain, and that the Floer homology of the two deformed curves recovers $I^\natural(K)$ [CHKK22, Conj. D]. In the algebraic model of §2.3 the assigned objects are considered up to homotopy equivalence.

2.2. Smith's coordinates and the sign of the perturbation. We use the coordinates of Smith [Smi24, §2.4], in which the boundary meridians satisfy $ac^{-1} = bd^{-1}$ and are sent to

$$a \mapsto \mathbf{i}, \quad b \mapsto e^{\gamma \mathbf{k}} \mathbf{i}, \quad c \mapsto e^{\theta \mathbf{k}} \mathbf{i}, \quad d \mapsto e^{(\gamma+\theta) \mathbf{k}} \mathbf{i}. \quad (2.1)$$

The letters a, b, c, d , and x, y in Remark 3.2, denote meridians in Smith's parametrization; we use them in this sense only in §2.1, here, in Remark 3.2 and in the proof of Proposition 3.4. Elsewhere b denotes a Maurer–Cartan element and c the simple closed curve of §4.2.

In these coordinates, for an appropriate holonomy perturbation, no perturbed tangle character variety meets the corner (π, π) . This is [CHKK22, Prop. 11.1], which is stated in coordinates in which the excluded corner is $[\gamma, \theta] = [0, \pi]$; the explicit curves of Sections 3 and 4 miss (π, π) by inspection (§3.3.2). Its proof applies verbatim: an abelian traceless representation is determined by how the tangle joins the four boundary points, and in Smith's coordinates the three possible joinings give the corner pairs $\{(0, 0), (0, \pi)\}$, $\{(0, 0), (\pi, 0)\}$ and $\{(\pi, 0), (0, \pi)\}$. We write Q_r for the rational tangle whose traceless character variety is the straight interval of slope r [Smi24, Prop. 2.27]. For $r = m/n$ in lowest terms, the line of Q_r joins $(\pi, 0)$ to $(0, \pi)$ if m and n are odd, $(0, 0)$ to $(0, \pi)$ if n is even, and $(0, 0)$ to $(\pi, 0)$ if m is even.

For tangles T_1, T_2 with character varieties in the pillowcase, Smith describes the character variety of the Conway sum $T_1 + T_2$ as a fiber product over γ : the θ -coordinates add, and over each *edge* $\gamma \in \{0, \pi\}$ a circle of representations appears when both summands are non-abelian there [Smi24, Thms. 3.15 and 3.26]. Smith calls such a circle *internal* when it misses the corners, and a *corner circle* otherwise [Smi24, Def. 3.24]. He perturbs the sum by a holonomy perturbation in a three-boundary configuration C_3 : the summands are glued along two inner boundary spheres, with pillowcases P_1 and P_2 , and the sum is read off on the outer sphere, with pillowcase P_3 . The perturbation depends on a real parameter t . For small $t > 0$, after an arbitrarily small additional perturbation where needed, each internal circle is replaced by two arcs crossing once, joined crosswise to the rest of the curve [Smi24, Thm. 4.22; the crossing is drawn in Figs. 3b and 23c]; we call this replacement the *cross-resolution* of the circle. By the symmetry described next the same holds for small $t < 0$ (Proposition 3.1). We write $\mathcal{R}_t(T_1 + T_2)$ for the perturbed character variety of the sum, mapped to P_3 .

The two signs of t give different curves. For odd q , a symmetry of Smith's equations maps $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ homeomorphically onto $\mathcal{R}_{-t}(Q_{1/3} + Q_{1/q})$ and acts on P_3 by the translation $(\gamma, \theta) \mapsto (\gamma + \pi, \theta)$ (Remark 3.2). This translation permutes the corners, and pairing dimensions with a fixed curve can change under it. *Throughout, $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ denotes Smith's perturbed curve for small $t < 0$, and every pairing dimension computed in Sections 3 and 4 is computed with this curve.* Remarks 3.17 and 4.3 record what changes for $t > 0$.

2.3. Twisted complexes over the algebra $\mathcal{B}$. Since tangle character varieties miss one corner, Cazassus, Herald, Kirk and Kotelskiy study them in the wrapped Fukaya category of the pillowcase with its four corners removed, restricted to curves that do not end at that corner, and generate the relevant subcategory by two arcs $a^\circ, a^\bullet$ ending at the other corners [CHKK22, §11.5]. The endomorphism algebra of $a^\circ \oplus a^\bullet$ is modelled by the algebra $\mathcal{B}$ of Kotelskiy, Watson and Zibrowius [KWZ19], an associative algebra whose two idempotents $\iota_\circ$ and $\iota_\bullet$ correspond to a° and $a^\bullet$.

To encode curves we use, following [KWZ19], the dual pair of arcs $s^\circ, s^\bullet$ in the chart obtained by sending a distinguished corner to infinity (Figure 1). Both are properly embedded with both ends at the puncture at infinity, and we call their union the *arc system*. Its complement has three faces, each containing one of the remaining corners, and we write L, M, R for the length-one paths that cross them. A positive path is a word X^k with $X \in \{L, M, R\}$ and $k \geq 1$, and we write $i \xrightarrow{X^k} j$ for a matrix entry from generator i to generator j labelled by X^k . Same-face words

compose by $X^m X^n = X^{m+n}$, products of words with two different face labels vanish, and the idempotents are units; the relations $DS = SD = 0$ of the quiver presentation of $\mathcal{B}$ are built into this notation. In that presentation [KWZ19, Def. 4.19], the words L^k and R^k are the powers of the loops D at the two vertices and the words M^k are the paths of length k in the two arrows S , as in the decomposition of $\mathcal{B}$ into one cyclic quiver for each face [KWZ19, Exs. 5.5 and 5.6]. The words L^k and R^k join generators of equal idempotent, and M^k joins generators of different idempotents exactly when k is odd. The algebra $\mathcal{B}$ is bigraded [KWZ19]. We ignore the gradings, as in the comparison in [CHKK22, §11.5], except in Section 4, where bigradings, with (π, π) as the special puncture, enter as a hypothesis of [KWZ19, Thm. 5.25], and in the observations of §3.3.3, Remark 3.16 and §5.2, which no proof uses.

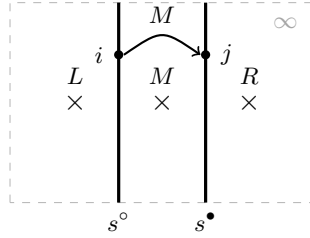


FIGURE 1. The chart for the algebra $\mathcal{B}$. A distinguished corner of the pillowcase is sent to infinity, leaving the other three corners ($\times$) in the plane. The arcs $s^\circ, s^\bullet$ of the arc system cut the chart into three faces, one for each remaining corner, labelled L , M and R . In the chart used throughout (§3.3.2) the corner at infinity is $(0, \pi)$, and the faces L , M and R contain the corners (π, π) , $(\pi, 0)$ and $(0, 0)$; Section 4 compares with the invariants of [KWZ19] in the normalization in which (π, π) is the special puncture (§4.1). A generator is an intersection of a curve with s° or $s^\bullet$ and carries the idempotent $\iota_\circ$ or $\iota_\bullet$; a segment running from i to j through the face X contributes an arrow $i \xrightarrow{X^k} j$, where k counts the crossings of the arcs. Each of the faces L and R is bounded by a single arc of the system, so their words join generators of equal idempotent; the face M is bounded by both arcs.

An immersed curve in the chart that misses the corners determines an object of the category $\text{Tw}(\mathcal{B})$ of twisted complexes in the standard way [KWZ19]: its intersections with the arc system are the generators, each carrying $\iota_\circ$ or $\iota_\bullet$ according to the arc it meets, and a segment of the curve running from one intersection to the next through the face X contributes the arrow labelled by the corresponding power of X . Each pillowcase intersection has two lifts to the torus double cover, interchanged by the elliptic involution ρ , so intersection counts on the torus are twice those on the pillowcase.

A finite twisted complex is a finite idempotent module V with a matrix δ over $\mathcal{B}$ satisfying $\delta^2 = 0$; it is a type D structure over $\mathcal{B}$ in the sense of [KWZ19], and we use the two descriptions interchangeably. An entry of δ labelled by an idempotent is a unit, and Gaussian elimination along it deletes its source and its target without changing the homotopy type; we call this a *unit cancellation*. A twisted complex is *reduced* if every entry of δ is a sum of positive words. Two reduced twisted complexes are *strictly isomorphic* if an idempotent-preserving bijection of their generators carries one differential to the other entry by entry. A reduced twisted complex is of *loop type* if every generator meets at most two arrows; each component of such a complex is then described by its *word*, the sequence of idempotents, labels and directions of its arrows read along

the component, and a strict isomorphism of loop-type complexes is decided by comparing the words of their components up to rotation and reversal.

For finite twisted complexes $X = (V_X, \delta_X)$ and $Y = (V_Y, \delta_Y)$, the morphism space $\text{Mor}(X, Y)$ consists of the matrices f over $\mathcal{B}$ from V_X to V_Y , with differential $f \mapsto \delta_Y f + f \delta_X$. We write $HF(X, Y) := H \text{Mor}(X, Y)$, and we call $\dim_{\mathbb{F}_2} HF(X, Y)$, when it is finite, the *pairing dimension* of X with Y . For curves in the position required by [KWZ19, Thm. 5.22], Kotelskiy, Watson and Zibrowius show that $\text{Mor}(X, Y)$ is quasi-isomorphic to the Lagrangian Floer complex of the two curves, so the pairing dimension is a Floer homology dimension. The hypotheses under which this identification is used are stated in Section 4.

The endomorphism differential of (V, δ) is $D_\delta(b) = \delta b + b \delta$. A matrix b over $\mathcal{B}$ is a *Maurer–Cartan element* for (V, δ) if

$$(\delta + b)^2 = D_\delta(b) + b^2 = 0, \quad (2.2)$$

and then $(V, \delta + b)$ is again a twisted complex. There is no infinite series in (2.2): for a suitable choice of Hamiltonian perturbations of the generating arcs, μ^1 and all μ^k with $k \geq 3$ vanish and μ^2 is concatenation of paths [CHKK22, §11.5]. Cazassus, Herald, Kirk and Kotelskiy deduce this from Bocklandt’s description of the wrapped Fukaya category of a punctured surface [Boc16, Thm. 7.6] and the constructions of [HKK17, §3]; the vanishing of the higher operations for a formal arc system is in [HKK17, §3.4]. The Maurer–Cartan elements used in Section 3 are supported near self-intersection points of a curve, where they encode smoothings of the curve.

Kotelskiy, Watson and Zibrowius associate to a four-ended tangle T in a ball type D structures $\widetilde{\text{BN}}(T)$ and $\widetilde{Kh}(T)$ over $\mathcal{B}$, invariants of T up to homotopy equivalence, with $\widetilde{Kh}(T)$ the mapping cone of $H \cdot \text{id}$ on $\widetilde{\text{BN}}(T)$ [KWZ19, Def. 6.5]; both are represented by immersed curves, possibly with local systems, in the four-punctured sphere [KWZ19]. Their geometric pairing theorem [KWZ19, Thm. 1.9] computes reduced Khovanov homology of a link $L = T_1 \cup T_2$ obtained by gluing two such tangles as

$$\widetilde{Kh}(L; \mathbb{F}_2) \cong HF(\widetilde{Kh}(mT_1), \widetilde{\text{BN}}(T_2)),$$

where m denotes the mirror. For rational tangles these curves are the perturbed traceless character varieties, $\widetilde{\text{BN}}$ corresponding to the tangle and $\widetilde{Kh}$ to the tangle with an earring [KWZ19, §1.8; for $\widetilde{\text{BN}}$ also §8.2, Prop. 8.3], up to the identification of charts made in Section 4. Section 4 compares Smith’s curve, with a Maurer–Cartan element, to $\widetilde{\text{BN}}(Q_{1/3} + Q_{1/q})$.

2.4. The knots. For odd $q \geq 3$ the pretzel knot $P(-2, 3, q)$ is the numerator closure

$$P(-2, 3, q) = \text{num}(Q_{-1/2} + Q_{1/3} + Q_{1/q}) \quad (2.3)$$

of a sum of rational tangles, in the convention of §2.2; we write $K_q = P(-2, 3, q)$. Its first two members are the torus knots $P(-2, 3, 3) = T(3, 4)$ and $P(-2, 3, 5) = T(3, 5)$, and for $q \geq 7$ it is hyperbolic [Wue26, Prop. 4.5]. Section 3 uses the decomposition (2.3) with the earring on $Q_{-1/2}$. The companion paper [Wue26] uses a different decomposition, obtained from Hedden, Herald and Kirk’s decomposition of $T(3, 5)$ by regluing, in which no bounding cochain is needed.

3. SMITH’S DECOMPOSITION

This section treats Smith’s decomposition (2.3) of K_q , with the earring on $Q_{-1/2}$: the piecewise-linear model of Smith’s curve (§3.1), its pairing dimension with zero bounding cochain at $q = 7$ (§3.3), and the Maurer–Cartan elements selected by a second closure (§3.4). Sections 3.2 and 3.5, on the triple point of a Lagrangian correspondence and on a Yoneda reduction, concern the formal route to identifying the assigned object and can be skipped on a first reading.

3.1. Piecewise-linear models and Smith’s cross-resolution. We work in Smith’s coordinates (2.1), with the corner rule for rational lines and the fiber-product description of Conway sums recalled in §2.2. As Smith does, we call the images of the lines $\gamma = 0$ and $\gamma = \pi$ the *edges* of the pillowcase, and the circles of representations that a Conway sum acquires over them *edge circles* [Smi24, Thms. 3.15 and 3.26]. Smith’s perturbation performs the cross-resolution (§2.2) of each internal edge circle [Smi24, Thm. 4.22]; we call the two arcs crossing once that replace the circle *connectors*, and the remaining arcs of the resolved curve, which come from the fiber product, *inherited strands*. The earring curve E_r of $\widehat{Q}_r$ is a Herald–Kirk figure-eight, the doubled arc closing around the two end corners, with one pinch [HK24, Def. 6.7]; this specifies the immersed curve only, and its bounding cochain is discussed after Proposition 3.7. In Smith’s notation $\widehat{Q}_r$ is the hat of Q_r (the tangle glued into the cobordism of [Smi24, Fig. 11a], with curve the reflection of the line of Q_r by $(\gamma, \theta) \mapsto (\gamma, -\theta)$ [Smi24, Lem. 2.31]), the earring is his $\natural$, and $\mathcal{R}^\natural$ denotes his perturbed character variety of a tangle carrying the earring, so that $E_r = \mathcal{R}^\natural(\widehat{Q}_r)$. The closing tangle enters the pairing through its mirror image, so E_r surrounds the line of Q_{-r} (§4.1); for instance $E_{-1/2} = \mathcal{R}^\natural(\widehat{Q}_{-1/2})$ lies along the line of slope $1/2$.

Write $L_{q,\text{PL}}$ for the piecewise-linear curve obtained from the lines of slopes $1/3$ and $1/q$ through $(0, 0)$ by the fiber product and the crosswise joining at every edge circle. These lines are the translates by $(0, \pi)$ of the lines of $Q_{1/3}$ and $Q_{1/q}$, and since the fiber product adds the θ -coordinates the two shifts cancel: both fiber products have the same image and the same edge circles. The model represents Smith’s perturbed curve for negative values of his perturbation parameter (Remark 3.2). The earring curve $E_{-1/2}$ is one immersed circle with one pinch; the model $L_{q,\text{PL}}$ has $q - 1$ resolved edge circles and is a single proper immersed arc. It has 40, 82 and 240 transverse self-intersections at $q = 5, 7, 11$. These are counts for the model: the number is not invariant under regular homotopy, and a numerical trace of Smith’s perturbed curve at $q = 7$ has 50 (Remark 3.15). At $q = 7$ the model also has twelve tangential self-contacts on the edges, where connectors meet inherited strands without crossing; we remove them by a small perturbation that creates no crossings. We require $\gcd(q, 3) = 1$, since otherwise corner circles appear; this excludes $q = 3, 9, 15, \dots$ from the members computed here, but not from [Wue26, Thm. 1.1].

Proposition 3.1 (Smith’s cross-resolution and the model). *Let q be odd and $\gcd(q, 3) = 1$.*

- (a) *The fiber product of the rational lines of slopes $1/3$ and $1/q$ has $q - 1$ edge circles, all internal, and no corner circles.*
- (b) *For all sufficiently small nonzero values t of Smith’s C_3 -perturbation parameter, after an arbitrarily small additional perturbation where Smith’s construction requires one, $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ contains a subspace homeomorphic to the curve obtained from the fiber product by replacing each internal circle with two intervals joined to its ends crosswise; this is the crosswise joining used to define $L_{q,\text{PL}}$, and the components of this subspace are the main components. As $t \rightarrow 0$ the image of each new interval approaches the image of the corresponding circle. The remaining components are circles, the auxiliary components.*
- (c) *For a fixed complementary tangle, there is an arbitrarily small perturbation of it for which the Lagrangian Floer homology of its curve with $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ equals that with the main components. This remains valid when either tangle carries an earring.*

Proof. Smith identifies the two rational character varieties with straight intervals of slopes $1/3$ and $1/q$ [Smi24, Prop. 2.27], and rational tangles of finite slope form a good pair, that is, satisfy Smith’s transversality conditions for a sum [Smi24, Def. 3.16], without preliminary perturbations [Smi24, example following Thm. 3.26]. Because the denominators 3 and q are coprime, the sum

has no corner circles [Smi24, discussion following Prop. 5.8], so every edge circle is internal. This is (a). For $t > 0$, part (b) is Smith's cross-resolution theorem [Smi24, Thm. 4.22], together with his remark preceding [Smi24, Thm. 4.20] that an arbitrarily small additional perturbation makes the relevant spaces manifolds when transversality fails. For $t < 0$ it follows from the case of $-t$ by the homeomorphism $\hat{\mathfrak{s}}$ of Remark 3.2, which maps $\mathcal{R}_{-t}(Q_{1/3} + Q_{1/q})$ onto $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ and the unperturbed fiber product onto itself. Part (c) is Smith's auxiliary-component result [Smi24, Prop. 5.7], which includes the earring case; its proof uses only that the auxiliary components converge to finitely many points of P^* as $t \rightarrow 0$, and by the same homeomorphism this also holds for $t < 0$. $\square$

Smith states informally that [Smi24, Thm. 4.22] determines the image of the perturbed curve up to regular homotopy ([Smi24, caption of Fig. 3] and the outline of the proof in his introduction); the formal statement gives the homeomorphism type and the convergence in (b). We use $L_{q,\text{PL}}$ as a representative of the regular-homotopy class of the main components, but no proof below depends on this identification; at $q = 7$ the type-D encoding of a traced copy of Smith's curve is strictly isomorphic to that of $L_{7,\text{PL}}$ (Remark 3.15).

Remark 3.2 (The sign of the perturbation). For the configuration C_3 of §2.2, write $C_{3,t}$ for its perturbed character variety. Its representations are parametrized by the images of the meridians a, b, c of the outer sphere and of a meridian x of the first inner sphere: the perturbation element is $p = \exp(t \operatorname{Im}(ac^{-1}x))$, the remaining meridians are $d = ca^{-1}b$ and $y = xp^{-1}a^{-1}pb$, and the perturbed variety is cut out by $\operatorname{Re}(y) = 0$ [Smi24, Prop. 4.1 and (4.1.1)]. The substitution $\mathfrak{s}: (x, t) \mapsto (-x, -t)$ leaves p and d unchanged and replaces y by $-y$, so it maps the perturbed variety for t onto that for $-t$. It is the identity on P_3 , and a direct computation with Smith's boundary words [Smi24, Fig. 20] shows that it acts on P_1 and on P_2 by the translation $(\gamma, \theta) \mapsto (\gamma, \theta + \pi)$. For odd q a line of slope $1/q$ is invariant under translation by (π, π) , so translation by $(0, \pi)$ carries the line of slope $1/q$ through $(0, 0)$ to the line through $(\pi, 0)$ and $(0, \pi)$. Consequently $\mathfrak{s}$ carries the composition of $C_{3,t}$ with the lines of slopes $1/3$ and $1/q$ through $(0, 0)$, for $t > 0$, onto the composition of $C_{3,-t}$ with the lines of $Q_{1/3}$ and $Q_{1/q}$, without moving the image in P_3 . Hence the piecewise-linear model, which is built from the lines through $(0, 0)$ and compared with their composition for $t > 0$, represents the main component of $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ in Smith's coordinates for small $t < 0$, which is the convention of §2.2.

The substitution $b \mapsto -b$ is also a symmetry of Smith's equations for each t , since p does not involve b and y changes sign; it translates each of P_1, P_2, P_3 by $(\gamma, \theta) \mapsto (\gamma + \pi, \theta)$. The composite $\hat{\mathfrak{s}}: (b, x, t) \mapsto (-b, -x, -t)$ therefore translates P_1 and P_2 by (π, π) , which preserves the lines of $Q_{1/3}$ and $Q_{1/q}$, and P_3 by $(\pi, 0)$, so the curve for $t > 0$ is the translate by $(\pi, 0)$ of the curve for $-t$. This translation is a symplectomorphism of P^* that permutes the corners; its effect on pairing dimensions is recorded in Remark 3.17.

3.2. The $Q_{1/3}$ correspondence is not embedded. Write P_i^* for the smooth locus of P_i , P^- for a pillowcase with the sign of its symplectic form reversed, and $C_{3,t}^{\text{reg}} \looparrowright P_1^- \times P_2^- \times P_3$ for the regular locus of the perturbed correspondence of the three-boundary configuration C_3 , Smith's $\mathcal{R}_{D_t}(C_3)^*$ [Smi24, Def. 4.9], which maps to the product of pillowcases by a Lagrangian immersion [Smi24, Cor. 4.11].

Lemma 3.3 (Reduction to a correspondence between surfaces). *Let $A_{1/3} \looparrowright P_1$ be the rational line of $Q_{1/3}$. Wherever the fiber product is transverse,*

$$F_{1/3,t} := A_{1/3} \times_{P_1} C_{3,t}^{\text{reg}} \looparrowright P_2^- \times P_3$$

is an immersed Lagrangian surface. On a torus-cover chart, with $\gamma \notin \{0, \pi\}$, of Smith's unperturbed manifold $R(C_3)^$ [Smi24, Def. 3.6] (on such a chart all representations are binary*

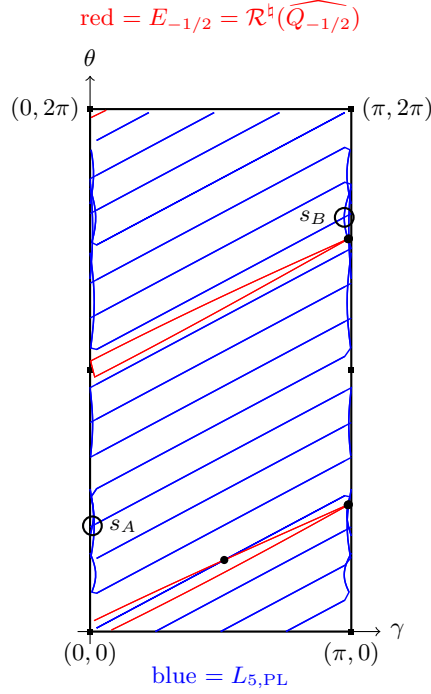


FIGURE 2. The two piecewise-linear model curves at $q = 5$ on the fundamental domain $[0, \pi] \times [0, 2\pi)$ of the pillowcase, in Smith’s coordinates (compare [Smi24, Fig. 31]). Blue is the model $L_{5,\text{PL}}$ of Smith’s perturbed curve $\mathcal{R}_t(Q_{1/3} + Q_{1/5})$, a single immersed arc with 40 self-intersections clustered along the edges; the small arcs hugging the edges $\gamma \in \{0, \pi\}$ are the connectors resolving the four edge circles, drawn at amplitude $\varepsilon \approx 0.05\text{--}0.07$, and the V-shaped turnbacks at the border are strands reflecting off the edges in the quotient. Red is the earring figure-eight $E_{-1/2} = \mathcal{R}^h(\widehat{Q_{-1/2}})$. Across the interior the blue strands run nearly parallel to red (slopes $8/15$ and $1/2$, crossing angle ≈ 0.026). Solid dots are the nine intersection points of the two curves: one interior, and two clusters of four near $\gamma = \pi$ that merge at this scale. The circled self-intersections s_A, s_B are the support of Computation A.2(i). Curves appear as several arcs because the fundamental domain folds the double cover.

dihedral), a branch of $A_{1/3}$ has $\theta_1 = \gamma/3 + \nu$ for a constant ν , and $F_{1/3,0}$ is the graph of the symplectic shear $(\gamma, \eta) \mapsto (\gamma, \eta + \gamma/3 + \nu)$; on compact subsets of such a chart the projection of $F_{1/3,t}$ to P_2 remains a local diffeomorphism for small $|t|$. After composition with the rational line of $Q_{1/7}$, the complement of these graph charts lies in neighborhoods of the six internal edge circles of Proposition 3.1 and of the two corner endpoints of the composed arc. If the fiber product is transverse along the entire regular locus, the target projection $F_{1/3,t} \rightarrow P_3^*$ is proper.

Proof. Over such a chart Smith’s boundary-coordinate formulas give $p_1 = (\gamma, \beta)$, $p_2 = (\gamma, \theta - \beta)$ and $p_3 = (\gamma, \theta)$ [Smi24, (3.1.1)–(3.1.6) and the proof of Prop. 3.20]. Imposing $\beta = \gamma/3 + \nu$ and writing $\eta = \theta - \beta$ gives the displayed graph, which preserves $d\gamma \wedge d\theta$, so the transverse composition is Lagrangian; the local-diffeomorphism property persists on compact subsets by the implicit function theorem. For $Q_{1/3}$ and $Q_{1/7}$ there are six internal edge circles, and [Smi24, Thm. 4.22] replaces precisely their neighborhoods by their cross-resolutions, as in the construction of $L_{7,\text{PL}}$. For properness, let $\overline{A}_{1/3}$ be the compact rational arc and $\overline{F}_{1/3,t} := \overline{A}_{1/3} \times_{P_1} \mathcal{R}_{D_t}(C_3)$, a

compact space; here $\mathcal{R}_{D_t}(C_3)$ is Smith's notation for $C_{3,t}$ [Smi24, §4.1]. The boundary restriction of $\mathcal{R}_{D_t}(C_3)$ lies either in $P_1^* \times P_2^* \times P_3^*$ or in the product of the three corner strata [Smi24, Prop. 4.3] (for $t < 0$ via Remark 3.2), and its regular Lagrangian locus is the former [Smi24, Cor. 4.11]. Hence $\overline{F}_{1/3,t} \setminus F_{1/3,t}$ maps to the corners of P_3 , and the inverse image of a compact subset of P_3^* is compact and lies in $F_{1/3,t}$. $\square$

Proposition 3.4 (A triple-point obstruction). *For every sufficiently small $t < 0$, the partial fiber product of Lemma 3.3 has three distinct points, at each of which the fiber product is transverse, with common image $((0, \pi/2), (0, \pi/2))$ in $P_2^* \times P_3^*$. Consequently $F_{1/3,t} \rightarrow P_2^- \times P_3$ is not embedded.*

Proof. Put $v(\varphi) = \cos \varphi \mathbf{i} + \sin \varphi \mathbf{k}$ and fix $\gamma = 0$, $\theta = \pi/2$ in Smith's parametrization, so that $a = b = \mathbf{i}$ and $c = d = \mathbf{j}$. For $\beta = 0$ or π write $(\varphi, u) = (\pi/2 - \alpha, t \sin \alpha)$ if $\beta = 0$ and $(\varphi, u) = (\pi/2 + \alpha, -t \sin \alpha)$ if $\beta = \pi$. Then $x = v(\varphi)$, the perturbation element is $p = e^{-u\mathbf{j}}$, and direct quaternion multiplication shows that $y = xp^{-1}a^{-1}pb$ is purely imaginary, so $\operatorname{Re}(y) = 0$ holds identically along these two families. On the torus cover the first boundary value is $(\gamma_1, \theta_1) = (2u, \varphi + 2u)$.

Take first $t > 0$ and the line $A_{1/3}^0 = \{3\theta_1 - \gamma_1 \in 2\pi\mathbb{Z}\}$ through $(0, 0)$, the translate of $A_{1/3}$ by $(0, \pi)$. Its equation $3\varphi + 4u \in 2\pi\mathbb{Z}$ gives

$$\begin{aligned} 3(\pi/2 - \alpha) + 4t \sin \alpha &= 0, & \alpha &\rightarrow \pi/2, & \beta &= 0, \\ 3(\pi/2 + \alpha) - 4t \sin \alpha &= 2\pi, & \alpha &\rightarrow \pi/6, & \beta &= \pi, \\ 3(\pi/2 + \alpha) - 4t \sin \alpha &= 4\pi, & \alpha &\rightarrow 5\pi/6, & \beta &= \pi, \end{aligned}$$

with α -derivatives $-3, 3, 3$ at $t = 0$, so the implicit function theorem gives three solutions for small $t > 0$. They are distinct, since their limiting x -values are, and none lies over a corner of P_1 , since $|\gamma_1| = 2t \sin \alpha > 0$. Using $xp = p^{-1}x$, the second-boundary words become $(v(\varphi + 2u), v(\varphi + 2u), \mathbf{j}, \mathbf{j})$, so $p_2 = (0, \pi/2)$ at all three points, and $(a, b, c, d) = (\mathbf{i}, \mathbf{i}, \mathbf{j}, \mathbf{j})$ gives $p_3 = (0, \pi/2)$. Since $\partial_\beta \operatorname{Re}(y) = -2t \sin^2 \alpha + O(t^2)$ and $\partial_\alpha(3\theta_1 - \gamma_1) = \pm 3 + O(t)$, the fiber product is transverse at all three points. Finally, the symmetry $\mathfrak{s}$ of Remark 3.2 maps the fiber product of $A_{1/3}^0$ with $C_{3,t}$ onto that of $A_{1/3}$ with $C_{3,-t}$, preserving distinctness, transversality and γ_1 , fixing p_3 and translating p_2 by $(0, \pi)$. Since $(0, 3\pi/2) = \rho(0, \pi/2)$, the three image points again map to $((0, \pi/2), (0, \pi/2))$; replacing t by $-t$ gives the statement. $\square$

Gao's wrapped representability theorem [Gao17, Thm. 7.24 and Cor. 7.25] applies to *admissible* correspondences, which are properly embedded, exact and cylindrical, with vanishing relative first Chern class [Gao17, Def. 6.3, Def. 7.2 and the standing convention of §7.1], and whose projection to the target is proper; only the geometric composition may be immersed [Gao17, Assumption 7.10], and the unobstructedness input [Gao17, Thm. 7.12] has the same hypotheses. By Proposition 3.4, Gao's theorem cannot be invoked for $F_{1/3,t}$, whose exactness, cylindricity and Chern class have not been checked either. The triple point is not generic, although the three sheets are pairwise transverse; splitting it into double points leaves $F_{1/3,t}$ non-embedded. Gao expects the results of his §6 to extend to immersed Lagrangians [Gao17, Rem. 6.1], but no immersed version of [Gao17, Thm. 7.24] is stated. Fukaya's compact theorem [Fuk25] does not apply verbatim to P^* , and the torus double cover does not help by itself: Smith's perturbed C_3 variety has singular points with neighborhoods that are cones on tori [Smi24, Thm. 4.16], and the required equivariant correspondence functor has not been constructed [CHKK22, §11.5].

3.3. The pairing dimension with zero bounding cochain at $q = 7$. We use the algebra $\mathcal{B}$, the arc system $s^\circ, s^\bullet$ with its faces L, M, R , the twisted complexes of curves, unit cancellations, morphism complexes and pairing dimensions as in §2.3, ignoring gradings, and we omit exponents

equal to one in arrows $i \xrightarrow{X^k} j$. A perturbation b of a twisted complex is Maurer–Cartan exactly when (2.2) holds, a finite identity. A recorded degree is the relative quantum grading gr_q of [KWZ19], computed in the chart below, that is, with $(0, \pi)$ as the special puncture; it plays no part in the Maurer–Cartan equation or in any pairing dimension.

3.3.1. Finiteness of the morphism complex. Since $\mathcal{B}$ is infinite dimensional, so is the morphism complex $\text{Mor}(\mathcal{E}, N)$ of two finite twisted complexes, and its homology need not be finite. The identity part $\text{Mor}^\times$ of $\text{Mor}(\mathcal{E}, N)$ is spanned by the entries labelled by the two vertex idempotents, and for $X \in \{L, M, R\}$ its X -part Mor^X by the entries labelled by positive powers of X . Put $U_L = L$, $U_R = R$ and $U_M = M^2$.

Lemma 3.5 (Mapping-cone reduction of the morphism complex). *Let $\mathcal{E}$ and N be reduced finite twisted complexes over $\mathcal{B}$.*

- (a) *The differential maps $\text{Mor}^\times$ into $\text{Mor}^+ := \text{Mor}^L \oplus \text{Mor}^M \oplus \text{Mor}^R$ and preserves each Mor^X . Hence $\text{Mor}(\mathcal{E}, N)$ is the mapping cone of a chain map $\beta: \text{Mor}^\times \rightarrow \text{Mor}^+$, where $\text{Mor}^\times$ is finite dimensional with zero differential.*
- (b) *Each Mor^X is a free $\mathbb{F}_2[U_X]$ -module of finite rank, with U_X acting by multiplication, and its differential is $\mathbb{F}_2[U_X]$ -linear. Consequently $H(\text{Mor}^X) \cong \mathbb{F}_2[U_X]^{f_X} \oplus \bigoplus_i \mathbb{F}_2[U_X]/(p_{X,i})$ for some $f_X \geq 0$ and nonconstant polynomials $p_{X,i}$.*
- (c) *$H\text{Mor}(\mathcal{E}, N)$ is finite dimensional if and only if $f_L = f_M = f_R = 0$, that is, if and only if each $\text{Mor}^X \otimes_{\mathbb{F}_2[U_X]} \mathbb{F}_2(U_X)$ is acyclic. In that case, with $e = \dim \text{Mor}^\times$ and $\tau = \dim_{\mathbb{F}_2} H(\text{Mor}^+) = \sum_{X,i} \deg p_{X,i}$, and with $\beta_*: \text{Mor}^\times \rightarrow H(\text{Mor}^+)$ the induced map,*

$$\dim_{\mathbb{F}_2} H\text{Mor}(\mathcal{E}, N) = e + \tau - 2 \text{rank } \beta_*. \quad (3.1)$$

Proof. Composing an idempotent entry with the positive words of $\delta_{\mathcal{E}}$ and δ_N gives positive words, and a positive power of X composed with a word in a different face is zero; this is (a). For (b), the entries of Mor^X joining a fixed pair of generators are the powers X^k , $k \geq k_0$, of a fixed parity when $X = M$; multiplication by U_X shifts k by 1 for $X = L, R$ and by 2 for $X = M$, so Mor^X is free on the entries of lowest length, and the differential commutes with U_X because powers of X commute. The decomposition is the structure theorem over the principal ideal domain $\mathbb{F}_2[U_X]$. For (c), the cone gives the exact sequence $0 \rightarrow \text{coker } \beta_* \rightarrow H\text{Mor}(\mathcal{E}, N) \rightarrow \ker \beta_* \rightarrow 0$, so the homology is finite exactly when $H(\text{Mor}^+)$ has no free summand, and f_X is the dimension of $H(\text{Mor}^X \otimes_{\mathbb{F}_2[U_X]} \mathbb{F}_2(U_X))$ by flatness. When $H(\text{Mor}^+)$ is finite, the sequence gives dimension $(\tau - \text{rank } \beta_*) + (e - \text{rank } \beta_*)$. $\square$

Remark 3.6. The mapping cone is the algebraic step preceding [KWZ19, Thm. 5.22]; Lemma 3.5 records the ungraded form used here. The torsion condition in (c) is a genuine hypothesis: for the one-generator complex $(\iota_\circ, 0)$ the endomorphism complex $\iota_\circ \mathcal{B} \iota_\circ$ has $f_L = f_M = 1$, and for the complex N below $\text{End}(N)$ has $f_M = f_R = 1$. For every pair of complexes whose morphism homology is used in this paper, exact computation over $\mathbb{F}_2[U_X]$ shows $f_L = f_M = f_R = 0$ and $p_{X,i} = U_X$ for all i , so that τ is the number of torsion summands.

3.3.2. The two explicit twisted complexes. We send the corner $(0, \pi)$ to infinity, so that the faces L , M and R contain the corners (π, π) , $(\pi, 0)$ and $(0, 0)$ (Figure 1). The curves encoded below, namely the arc N with ends at $(0, 0)$ and $(\pi, 0)$, its smoothings and closed earring curves, meet neither $(0, \pi)$ nor the corner (π, π) that no tangle character variety meets. The description of [CHKK22, §11.5] applies to any corner, since translations of the pillowcase permute the corners, and in either case morphisms are computed in the wrapped category of the pillowcase with all four corners removed, so the morphism homology does not depend on which of the two corners is

at infinity. For a slope r we write $\mathcal{E}_r$ for the reduced twisted complex that encodes the earring curve E_r in this chart.

In this chart the model $L_{7,\text{PL}}$ meets the arc system in 74 points of the torus cover, that is, in 37 points of the pillowcase. Three unit cancellations leave the 31-generator complex N whose $\iota_\circ$ generators are 1, 2, 5, 6, 15, 16, 20, 21, 32, 33; the remaining generators in

$$V_N = \{0, 1, \dots, 10, 13, 14, \dots, 23, 26, 27, 30, 31, \dots, 36\}$$

carry $\iota_\bullet$, and

$$\begin{aligned} \delta_N = & 17 \xrightarrow{R} 18 + 16 \xrightarrow{M} 17 + 16 \xrightarrow{L} 15 + 15 \xrightarrow{M} 14 + 14 \xrightarrow{R} 13 + 13 \xrightarrow{M^2} 10 + 10 \xrightarrow{R} 9 + 8 \xrightarrow{M^2} 9 \\ & + 7 \xrightarrow{R} 8 + 6 \xrightarrow{M} 7 + 6 \xrightarrow{L} 5 + 5 \xrightarrow{M} 4 + 4 \xrightarrow{R} 3 + 2 \xrightarrow{M} 3 + 2 \xrightarrow{L} 1 + 1 \xrightarrow{M} 0 \\ & + 0 \xrightarrow{R} 19 + 20 \xrightarrow{M} 19 + 20 \xrightarrow{L} 21 + 21 \xrightarrow{M} 22 + 22 \xrightarrow{R} 23 + 23 \xrightarrow{M^2} 26 + 26 \xrightarrow{R} 27 \\ & + 30 \xrightarrow{M^2} 27 + 31 \xrightarrow{R} 30 + 32 \xrightarrow{M} 31 + 32 \xrightarrow{L} 33 + 33 \xrightarrow{M} 34 + 34 \xrightarrow{R} 35 + 35 \xrightarrow{M^2} 36. \end{aligned}$$

Direct path multiplication gives $\delta_N^2 = 0$. The earring curve $E_{-1/2}$ meets the arc system in eight points, and one unit cancellation leaves a six-generator complex $\mathcal{E} = \mathcal{E}_{-1/2}$ with $\iota_\circ$ generators 5, 6, $\iota_\bullet$ generators 0, 3, 4, 7, and

$$\delta_{\mathcal{E}} = 3 \xrightarrow{M^2} 0 + 4 \xrightarrow{R} 3 + 5 \xrightarrow{M} 4 + 6 \xrightarrow{L} 5 + 6 \xrightarrow{M} 7 + 0 \xrightarrow{R} 7, \quad (3.2)$$

again with $\delta_{\mathcal{E}}^2 = 0$.

3.3.3. Smoothings and switches. A self-intersection of $L_{7,\text{PL}}$ is *connector-inherited* if one branch is a connector and the other an inherited strand, and *connector-connector* if both branches are connectors. The lift of $L_{7,\text{PL}}$ to the torus double cover is a single closed immersed curve C of homology class $(21, 10)$; the involution ρ reverses its orientation, and each self-intersection orbit s of the model consists of two transverse double points of C interchanged by ρ . We label the orbits $S_0, S_1, \dots$ in the order in which the ancillary files enumerate them (§A.5); the indices are arbitrary labels; the four orbits of Proposition 3.7 are identified by their coordinates, the others through the ancillary files. Exactly one of the two smoothings of a double point is compatible with the orientation of C ; it splits off the loop of C between the two visits. The *oriented smoothing* at s performs it at both points of s , in disjoint disks; the pair is ρ -invariant and defines a smoothing of the pillowcase curve. Performing the other smoothing at both points gives the *unoriented smoothing*. At 38 of the 82 orbits the two branches lie on the same lift of the pillowcase arc, and there the oriented smoothing is also compatible with an orientation of that arc; the four orbits below are of this kind. A smoothing is *generator-preserving* if, after the three unit cancellations, its encoding has the same idempotent module V_N ; its differential then differs from δ_N by an element $b_s \in \text{End}(N)$, the *switch* at s . An *algebraic switch* of a reduced twisted complex X is an element of $\text{End}(X)$ that replaces two arrows $i \xrightarrow{Y^k} j$ and $i' \xrightarrow{Y^k} j'$ of δ_X , with j and j' of the same idempotent, by $i \xrightarrow{Y^k} j'$ and $i' \xrightarrow{Y^k} j$, and for which the result satisfies the Maurer–Cartan equation. A switch is defined by a smoothing of the curve, whereas an algebraic switch is defined by the complex alone and need not come from a smoothing.

Proposition 3.7. *The pairing dimension of $\mathcal{E}$ with N is*

$$\dim_{\mathbb{F}_2} H \text{Mor}_{\text{Tw}(\mathcal{B})}(\mathcal{E}, N) = 7.$$

At each of the four self-intersection orbits $S_{18}, S_{25}, S_{69}, S_{74}$ of $L_{7,\text{PL}}$ tabulated below, the oriented smoothing produces a four-arrow switch $b_s \in \text{End}(N)$ with

$$D_N(b_s) = 0, \quad b_s^2 = 0, \quad \dim_{\mathbb{F}_2} H \text{Mor}_{\text{Tw}(\mathcal{B})}(\mathcal{E}, (N, b_s)) = 9.$$

Since the Maurer–Cartan equation in $\text{Tw}(\mathcal{B})$ is the finite identity $(\delta_N + b_s)^2 = 0$, the elements b_s satisfy the Maurer–Cartan equation with no higher terms. The four orbits and their switches are as follows; the last column records the multiset of quantum degrees gr_q of the four terms, relative to the grading of N , and is not used in the proof.

s	(γ, θ)	old arrows	new arrows	gr_q
S_{18}	(0.0304, 4.502)	$4 \xrightarrow{R} 3, 22 \xrightarrow{R} 23$	$22 \xrightarrow{R} 3, 4 \xrightarrow{R} 23$	$-8, 0, 0, 8$
S_{25}	(3.083, 1.767)	$6 \xrightarrow{L} 5, 20 \xrightarrow{L} 21$	$20 \xrightarrow{L} 5, 6 \xrightarrow{L} 21$	$-8, 0, 0, 8$
S_{69}	(0.0416, 5.405)	$0 \xrightarrow{R} 19, 26 \xrightarrow{R} 27$	$0 \xrightarrow{R} 27, 26 \xrightarrow{R} 19$	$-8, 0, 0, 8$
S_{74}	(3.080, 3.561)	$20 \xrightarrow{L} 21, 32 \xrightarrow{L} 33$	$20 \xrightarrow{L} 33, 32 \xrightarrow{L} 21$	$-4, 0, 0, 4$

Each smoothing is disjoint from the arc system, so it keeps the generators, and b_s is the sum of the two old and the two new arrows. Every displayed arrow has relative δ -degree -1 . The column gr_q uses $(0, \pi)$ as the special puncture, the normalization of the chart of §3.3.2, and in it none of the four deformations is bigraded. In the normalization of Section 4, in which (π, π) is special (§4.1), the column reads $-8, 0, 0, 8$; $-8, 0, 0, 8$; $0, 0, 0, 0$; $-4, 0, 0, 4$: the object $(N, b_{S_{69}})$ is bigraded there, and the other three are not. Gradings play no part in this section.

Proof of Proposition 3.7. For each row, every composite of an arrow of b_s with an arrow of δ_N , in either order, is zero in $\mathcal{B}$, because the face labels differ or the endpoints do not match; and $b_s^2 = 0$ because no target of an arrow of b_s is the source of another. Direct encoding of the smoothed curve gives $\delta_{N_s} = \delta_N + b_s$ entry by entry (Appendix A.4). The oriented smoothing splits C into three torus components: a ρ -invariant one, which projects to an arc, and two that project to one closed pillowcase component, the loop. With the orientations induced from C , their classes and the pairing dimensions of $\mathcal{E}$ with arc and loop are

s	torus component classes	pairing dimensions, arc + loop
S_{18}, S_{25}	$(17, 6), (2, 2), (2, 2)$	$5 + 4$
S_{69}	$(17, 8), (2, 1), (2, 1)$	$5 + 4$
S_{74}	$(15, 8), (3, 1), (3, 1)$	$7 + 2$

The morphism complexes are computed in full, without truncating path length. For each pair the torsion condition of Lemma 3.5(c) holds by exact computation over $\mathbb{F}_2[U_X]$, and (3.1) gives the dimensions from the following data:

object or component	e	τ	$\text{rank } \beta_*$	$e + \tau - 2 \text{rank } \beta_*$
N	104	105	101	7
$N_{S_{18}}, N_{S_{25}}$: arc; loop	80; 24	85; 20	80; 20	5; 4
$N_{S_{69}}$: arc; loop	84; 20	85; 20	82; 18	5; 4
$N_{S_{74}}$: arc; loop	76; 28	75; 30	72; 28	7; 2

Hence the undeformed pairing dimension is seven and the four deformed pairing dimensions are nine. $\square$

By [Wue26, Thm. 1.1], $\dim_{\mathbb{F}_2} I^{\natural}(K_7) = 9$. Under Conjecture D of [CHKK22] the earring curve $E_{-1/2}$ carries the zero cochain: Smith proves this for the earring on Q_0 by comparison with the two-component unlink [Smi24, Lem. 5.3], and the argument applies verbatim to $\widehat{Q}_r$ for every slope r , with the unlink $\text{num}(Q_r + Q_{-r})$ in place of $\text{num}(Q_0 + Q_0)$; Smith uses it for $r = -1/2$ [Smi24, proof of Thm. 5.9]. Conditional on [CHKK22, Conj. D], Proposition 3.7 therefore implies that the object assigned to $Q_{1/3} + Q_{1/7}$ is not N with the zero cochain.

3.4. Selection by a second closure. Pairing dimension nine alone does not determine the object: among the four deformations of Proposition 3.7 it is attained by three homotopy-equivalence classes (Corollary 3.9), and in the enumeration of Proposition 3.10 it is attained at 11 orbits by 9 distinct switches. We therefore pair the same tangle with a second closure, whose instanton homology is computed by the squeeze of [Wue26, §3].

Proposition 3.8 (The second closure). *Set $K_* := \text{num}(Q_{-3/4} + Q_{1/3} + Q_{1/7})$. Then $I^\natural(K_*; \mathbb{Z}) \cong \mathbb{Z}^{31}$. Encoding the piecewise-linear curve of the earring tangle $\widehat{Q_{-3/4}}$ as above gives the 14-generator twisted complex $\mathcal{E}_* = \mathcal{E}_{-3/4}$ of (3.3). Its pairing dimensions with (N, b_s) are 31, 31, 23 and 25 for $s = S_{18}, S_{25}, S_{69}$ and S_{74} , and 25 with N .*

Proof. The knot K_* has the 14-crossing planar diagram

$$\begin{aligned} &(24, 13, 25, 14), (10, 25, 11, 26), (26, 11, 27, 12), (12, 27, 13, 0), (7, 14, 8, 15), (15, 8, 16, 9), \\ &(9, 16, 10, 17), (23, 6, 24, 7), (5, 22, 6, 23), (21, 4, 22, 5), (3, 20, 4, 21), (19, 2, 20, 3), \\ &(1, 18, 2, 19), (17, 0, 18, 1) \end{aligned}$$

in the usual PD convention; in the knot table distributed with SnapPy [CDGW] it is $K14n12201$. Seifert's algorithm gives

$$\Delta_{K_*}(t) = t^{-6} - t^{-5} + 2t^{-3} - 4t^{-2} + 5t^{-1} - 5 + 5t - 4t^2 + 2t^3 - t^5 + t^6,$$

so the coefficient norm is $\|\Delta_{K_*}\| = 31$, and $|\Delta_{K_*}(-1)| = 23$. The reduced Khovanov homology of K_* over $\mathbb{Z}$ is free, with ranks $(1, 0, 1, 1, 1, 2, 3, 3, 4, 5, 4, 3, 2, 1)$ in homological degrees 0 through 13, summing to 31 (computed with Khoca [Lew26]). The proof of [Wue26, Thm. 1.1] applies verbatim. It squeezes $I^\natural$ between two bounds: the coefficient norm $\|\Delta_K\|$ of the Alexander polynomial, the sum of the absolute values of its coefficients, bounds $\text{rank } I^\natural(K)$ below [KM10], and the rank of reduced Khovanov homology bounds it above, by the spectral sequence of Kronheimer and Mrowka [KM11b]. For K_* the two bounds agree, the integral spectral sequence collapses on a free E_2 page, and $I^\natural(K_*; \mathbb{Z}) \cong \mathbb{Z}^{31}$. The knot K_* is not Khovanov-thin; the squeeze closes because $\|\Delta_{K_*}\|$ equals the Khovanov rank. The complex $\mathcal{E}_*$ has $\iota_\circ$ generators 1, 2, 7, 8, 11, 12, $\iota_\bullet$ generators 0, 3, 4, 5, 6, 9, 10, 13, and

$$\begin{aligned} \delta_{\mathcal{E}_*} = & 1 \xrightarrow{M} 0 + 1 \xrightarrow{L} 2 + 2 \xrightarrow{M} 3 + 3 \xrightarrow{R} 4 + 5 \xrightarrow{M^2} 4 + 6 \xrightarrow{R} 5 + 7 \xrightarrow{M} 6 \\ & + 8 \xrightarrow{L} 7 + 8 \xrightarrow{M} 9 + 10 \xrightarrow{R} 9 + 11 \xrightarrow{M} 10 + 12 \xrightarrow{L} 11 + 12 \xrightarrow{M} 13 + 0 \xrightarrow{R} 13. \end{aligned} \quad (3.3)$$

For each of the five pairs the torsion condition of Lemma 3.5(c) holds by exact computation, and the data $(e, \tau, \text{rank } \beta_*)$ are $(228, 251, 227)$ for N and for $(N, b_{S_{74}})$, $(228, 251, 224)$ for $(N, b_{S_{18}})$ and $(N, b_{S_{25}})$, and $(228, 251, 228)$ for $(N, b_{S_{69}})$, giving 25, 31 and 23. $\square$

Corollary 3.9. *The four deformations of Proposition 3.7 represent exactly three homotopy-equivalence classes in $\text{Tw}(\mathcal{B})$. The S_{18} and S_{25} objects are strictly isomorphic; their common class, the S_{69} class and the S_{74} class are pairwise distinct.*

Proof. The idempotent-preserving permutation $(4\ 22)(5\ 21)$, extended by the identity on the other 27 generators, carries $\delta_N + b_{S_{18}}$ entry by entry to $\delta_N + b_{S_{25}}$. Pairing with $\mathcal{E}_*$ is a dg functor, so $\dim H \text{Mor}(\mathcal{E}_*, -)$ is a homotopy invariant; by Proposition 3.8 it takes the values 31, 23 and 25 on the S_{18}/S_{25} , S_{69} and S_{74} objects. The proof does not use the classification of curves by twisted complexes [CHKK22, §11.5], [HKK17, Thm. 4.3], [KWZ19, Thm. 1.5]; the last two concern graded and bigraded objects. $\square$

Propositions 3.10–3.12 treat smoothings at all self-intersection orbits of $L_{7,\text{PL}}$, which, with their smoothings, are read off from the floating-point geometry of the model (Appendix A.4); the computations with the resulting complexes are exact over $\mathbb{F}_2$.

Proposition 3.10 (Oriented smoothings). *The model $L_{7,\text{PL}}$, with its self-intersections and their smoothings computed as in Appendix A.4, has 82 self-intersection orbits, 52 connector–inherited and 30 connector–connector. Of the 82 oriented smoothings, 73 are generator-preserving. They determine 45 distinct, linearly independent switches $b_s \in \text{End}(N)$, and every one satisfies $D_N(b_s) = 0$ and $b_s^2 = 0$. Among these 73 smoothings, the simultaneous pairing dimensions 9 with $\mathcal{E}$ and 31 with $\mathcal{E}_*$ occur only at S_{18} and S_{25} . These are also the only orbits at which the shorter branch loop has torus class $(2, 2)$ up to sign; here a double point of the orbit cuts C into two loops, and the shorter one is that whose torus class has smaller ℓ^1 -norm.*

Proof. The provenance of each segment partitions the 82 orbits into 52 connector–inherited and 30 connector–connector orbits. No orbit has two inherited branches: since the fiber product adds the θ -coordinates, the inherited strands are segments of lines of the common slope $1/3 + 1/7 = 10/21$ on the torus cover, and so are pairwise disjoint. The oriented smoothing is encoded directly at every orbit. The nine orbits of Proposition 3.11 change the cancelled module; at every other orbit the two differentials differ by two old and two new arrows, and exact path multiplication gives $D_N(b_s) = b_s^2 = 0$. Coinciding switches leave 45 distinct elements, linearly independent in the basis of arrows. The torsion condition of Lemma 3.5(c) holds for each pair by exact computation, and applying (3.1) with both earrings gives the following distribution of the pairs $(\dim H \text{Mor}(\mathcal{E}, (N, b_s)), \dim H \text{Mor}(\mathcal{E}_*, (N, b_s)))$:

pair	(5, 23)	(5, 25)	(7, 23)	(7, 25)	(7, 27)	(9, 23)	(9, 25)	(9, 31)
orbits	1	2	12	39	8	1	8	2
distinct switches	1	2	6	23	4	1	6	2

whose last column consists of S_{18} and S_{25} . For the class label, the two arcs of C between the two visits of a torus point of an orbit close up to complementary branch loops with classes summing to $(21, 10)$; the class of smaller ℓ^1 norm, up to sign, is the same at both torus points, and $(2, 2)$ occurs only at S_{18} and S_{25} . $\square$

Proposition 3.11 (The remaining single smoothings). *The oriented smoothings at the nine orbits $S_7, S_8, S_{23}, S_{24}, S_{49}, S_{52}, S_{53}, S_{61}, S_{73}$ are not generator-preserving. Their encodings N_s are 31-generator twisted complexes whose pairing dimensions with $(\mathcal{E}, \mathcal{E}_*)$ are $(3, 23)$ at S_7 , $(5, 23)$ at S_8, S_{53} and S_{73} , and $(7, 25)$ at the other five. Hence, among all 82 oriented smoothings, the pair $(9, 31)$ occurs only at S_{18} and S_{25} . Among the 82 unoriented smoothings the pair $(9, 31)$ also occurs exactly at S_{18} and S_{25} ; there the smoothed curve is connected, its encoding has 21 generators, and $\mathcal{E}_{3/4}$, the complex of the earring curve of $\widehat{Q_{3/4}}$, has pairing dimension 69 with it, against 103 with $(N, b_{S_{18}})$. The knot $\text{num}(Q_{3/4} + Q_{1/3} + Q_{1/7})$ is alternating with determinant 103, so $\dim_{\mathbb{Q}} I^{\natural} = 103$.*

Proof. The smoothed curves at the nine excluded orbits and the unoriented smoothings at all 82 orbits are encoded as standalone twisted complexes and paired with $\mathcal{E}$ and $\mathcal{E}_*$, and at S_{18} and S_{25} with $\mathcal{E}_{3/4}$, with the torsion condition verified in each case. The knot $\text{num}(Q_{3/4} + Q_{1/3} + Q_{1/7})$ is the numerator closure of three rational tangles of the same sign, hence alternating, with determinant $|4 \cdot 3 \cdot 7(3/4 + 1/3 + 1/7)| = 103$; alternating knots are quasi-alternating [OS05, Lemma 3.2], and for quasi-alternating knots the rank of $I^{\natural}$ is the determinant [KM11b, Cor. 1.6]. $\square$

Proposition 3.12 (How far the two closures reach). *Every element $b = \epsilon_{18}b_{S_{18}} + \epsilon_{25}b_{S_{25}} + \epsilon_{69}b_{S_{69}} + \epsilon_{74}b_{S_{74}}$, $\epsilon_s \in \mathbb{F}_2$, is Maurer–Cartan, and among these 16 deformations the pair $(9, 31)$ occurs only for $b = b_{S_{18}}$ and $b = b_{S_{25}}$. The 46 generator-preserving oriented smoothings at connector–inherited orbits give 38 distinct switches; these are linearly independent, every sum of two of them is Maurer–Cartan, and hence all 2^{38} elements of their span are distinct and*

Maurer–Cartan. Of the $\binom{38}{2} = 703$ two-switch sums, 41 also have the pair $(9, 31)$, and none of these 41 complexes is strictly isomorphic to $(N, b_{S_{18}})$.

Proof. Exact multiplication verifies the Maurer–Cartan equation for the sum of each pair of the 38 connector–inherited switches; since each switch is closed and square-zero, these checks say that all mixed anticommutators vanish, so the whole span is Maurer–Cartan. Applying (3.1) with both earrings to all 703 two-switch sums gives 41 with the pair $(9, 31)$. Regarding each differential as a directed graph on the 31 generators, labelled by idempotents, faces and path lengths, an exhaustive search over label-preserving bijections finds none carrying the S_{18} differential to any of the 41, and does find the S_{18} – S_{25} permutation; this excludes strict isomorphism only, not homotopy equivalence. Every sum of the four named switches is Maurer–Cartan; eight of the 16 have pairing dimension seven with $\mathcal{E}$, and the other eight have pairing dimension nine with $\mathcal{E}$ and pairing dimension with $\mathcal{E}_*$ equal to 31 for $b_{S_{18}}$ and $b_{S_{25}}$; 23 for $b_{S_{69}}$, $b_{S_{18}} + b_{S_{25}} + b_{S_{69}}$, $b_{S_{25}} + b_{S_{69}} + b_{S_{74}}$ and $b_{S_{18}} + b_{S_{25}} + b_{S_{69}} + b_{S_{74}}$; and 25 for $b_{S_{74}}$ and $b_{S_{18}} + b_{S_{74}}$. $\square$

Hypothesis 3.13 ($q = 7$ local support). *If the tangle assignment of Cazassus, Herald, Kirk and Kotelskiy is defined for $Q_{1/3} + Q_{1/7}$ with Smith’s perturbation for small $t < 0$, the object of $\text{Tw}(\mathcal{B})$ that it determines (an immersed curve with a bounding cochain) is homotopy equivalent to one of the following: (N, b_s) for one of the 45 distinct switches of Proposition 3.10; the encoding N_s of the oriented smoothing at one of the nine orbits of Proposition 3.11; or (N, b) for an element $b \in \text{span}_{\mathbb{F}_2}\{b_{S_{18}}, b_{S_{25}}, b_{S_{69}}, b_{S_{74}}\}$. Equivalently, it is homotopy equivalent to the encoding of the oriented smoothing at one of the 82 self-intersection orbits, or to (N, b) for b in this span.*

In the language of immersed Floer theory, apart from the span of the four switches, the hypothesis says that the bounding cochain is concentrated at one self-intersection of the model and acts there as the oriented smoothing. The span of the four switches of Proposition 3.7, each of which reaches pairing dimension nine, is included so that the hypothesis also covers combinations of these deformations. The hypothesis refers to the representative $L_{7, \text{PL}}$, perturbed off its tangential self-contacts; its set of self-intersections is not an invariant of the regular-homotopy class (Remark 3.15).

Corollary 3.14 (Conditional selection). *Assume Conjecture D of [CHKK22] and Hypothesis 3.13. Then the object assigned to $Q_{1/3} + Q_{1/7}$ is homotopy equivalent to the common S_{18}/S_{25} class.*

Proof. Under Conjecture D of [CHKK22] the cochains on $\widehat{Q_{-1/2}}$ and $\widehat{Q_{-3/4}}$ vanish, by the argument following Proposition 3.7, so pairing the assigned object with the two earrings must give the pairing dimensions 9 and 31, which are homotopy invariants. Among the objects allowed by Hypothesis 3.13, Propositions 3.10–3.12 leave only $(N, b_{S_{18}})$ and $(N, b_{S_{25}})$, which are strictly isomorphic by Corollary 3.9. $\square$

Remark 3.15 (The traced analytic curve). A numerical trace of Smith’s perturbed curve $\mathcal{R}_t(Q_{1/3} + Q_{1/7})$ at $t = -0.015$ and at $t = -0.02$, computed through the symmetry $\mathfrak{s}$ of Remark 3.2 from the composition of $C_{3, -t}$ with the lines of slopes $1/3$ and $1/7$ through $(0, 0)$, has 50 transverse self-intersection orbits rather than 82, and its type-D encoding is strictly isomorphic to N . On the traced curve 43 of the oriented smoothings are generator-preserving, with 43 distinct switches, and the pair $(9, 31)$ occurs at exactly two of them, the orbits whose shorter branch loop has class $(2, 2)$, which correspond to S_{18} and S_{25} . The oriented smoothings at the other seven orbits change the cancelled module, and their pairing dimensions were not computed. Among generator-preserving smoothings the selection is therefore not an artifact of the piecewise-linear model, although the trace is a floating-point computation and not a proof.

Remark 3.16 (Scope). Proposition 3.7 proves existence of four Maurer–Cartan elements, and Proposition 3.10 an exhaustive enumeration in the stated geometric class; neither proves selection

by the tangle. The δ -degree -1 part of the endomorphism cohomology of N is 20-dimensional, and its quantum-degree-zero part is one-dimensional in either normalization. With $(0, \pi)$ special it is spanned by the class of $b_{S_{10}}$, equivalently of the arrow $2 \xrightarrow{M} 9$, and $(N, b_{S_{10}})$ has pairing dimensions 5 and 25 with $\mathcal{E}$ and $\mathcal{E}_*$. With (π, π) special it is spanned by the class of $b_{S_{69}}$, and $(N, b_{S_{69}})$ has pairing dimensions 9 and 23. Corollary 3.14 assumes Conjecture D of [CHKK22] and that the assigned object lies in one of the stated classes, and the algebraic calculation proves neither assumption. The restriction on support is essential: Proposition 3.12 exhibits 41 two-switch sums outside the hypothesis with the same two closure dimensions. The four elements b_s of Proposition 3.7 are not expressed in the immersed Floer complex of Akaho and Joyce [AJ10], whose generators at a transverse self-intersection are its two ordered branch jumps. Among them only $b_{S_{69}}$ is homogeneous in the quantum grading, and only in the normalization of Section 4.

Remark 3.17 (The opposite sign of the perturbation). For small $t > 0$ the curve, its self-intersections and their smoothings are the translates by $(\gamma, \theta) \mapsto (\gamma + \pi, \theta)$ of those used here (Remark 3.2), while the earring curves do not depend on t ; since the translation is a symplectomorphism of P^* of order two, we computed the pairing dimensions for $t > 0$ as those of the translated earrings with the present objects. The undeformed pairing dimensions remain 7 and 25, and the smoothings at S_{18} and S_{25} still give $(9, 31)$, but those at S_{69} and S_{74} give $(5, 25)$ and $(7, 25)$. Pairing dimension nine with the first earring is reached by seven of the 45 distinct switches: those at S_{10} , S_{18} , S_{22} , S_{25} , S_{29} , S_{30} , and the one shared by S_{26} , S_{27} , S_{28} , S_{32} , S_{33} . Among the 82 oriented smoothings, the 82 unoriented smoothings and the 16 elements of the span of Proposition 3.12, the pair $(9, 31)$ again occurs only at S_{18} and S_{25} , the unoriented smoothings there are again separated by the third closure (69 against 103), and in the span only $b_{S_{18}}$ and $b_{S_{25}}$ reach pairing dimension nine. The two-closure selection therefore holds for either sign of t , whereas Proposition 3.7 does not: for $t > 0$ only two of the four objects have pairing dimension nine. All other values in this section are stated for $t < 0$; the numbers h_q^{big} of Table 1 are the same for both signs.

3.5. A two-generator Yoneda reduction. A representability theorem would reach the assigned object without identifying the virtual chain of every figure-eight bubble: Fukaya, in the compact setting [Fuk25, Thms. 6.3 and 7.3], and Gao, under the hypotheses recalled in §3.2, obtain the composed object with its bounding cochain by a cyclic element and the Yoneda lemma. Since N and the b_s are encoded in the chart of §3.3.2, which sends $(0, \pi)$ to infinity, let $\mathcal{C} \subset \mathcal{W}(P^*)$ be the full subcategory of [CHKK22, §11.5] of unobstructed objects missing the corner $(0, \pi)$ (the description there applies to any corner, §3.3.2), and let $\mathbf{a} = a^\circ \oplus a^\bullet$ be the sum of the arcs ending at the other corners, so that $\mathcal{B} = \text{End}_{\mathcal{C}}(\mathbf{a})$. Then $\mathbf{a}$ generates $\mathcal{C}$ and $\text{Tw } \mathcal{C} \simeq \text{Tw}(\mathcal{B})$ [CHKK22, §11.5], by [Boc16, Thm. 7.6] and [HKK17, §3].

Remark 3.18 (Yoneda recognition). Assume that the quilted module $\mathcal{M}_{F_{1/3,t}}(A_{1/7})$, the A_∞ module over $\mathcal{C}$ whose value on an object Y is the quilted Floer complex of Y , $F_{1/3,t}$ and $A_{1/7}$, is defined and is represented by an object $X_{1/3,1/7} \in \mathcal{C}$, and that there is an A_∞ quasi-isomorphism of right $\mathcal{B}$ -modules

$$\mathcal{M}_{F_{1/3,t}}(A_{1/7})|_{\mathcal{B}} \simeq \text{Mor}_{\text{Tw}(\mathcal{B})}(\mathbf{a}, (N, b_{S_{18}})).$$

Then $X_{1/3,1/7} \simeq (N, b_{S_{18}}) \simeq (N, b_{S_{25}})$ in $\text{Tw } \mathcal{C}$. This is the Yoneda lemma: since $\mathbf{a}$ generates $\mathcal{C}$, restricted Yoneda is cohomologically fully faithful on $\mathcal{C}$, with values in perfect $\mathcal{B}$ -modules; representability identifies the first module with the Yoneda module of $X_{1/3,1/7}$, so the assumed quasi-isomorphism is the image of a homotopy equivalence $X_{1/3,1/7} \simeq (N, b_{S_{18}})$, and the second equivalence is Corollary 3.9. The module must be given on a° and $a^\bullet$ as full chain-level complexes

with every $\mathcal{B}$ -action and higher module map; pairing dimensions with two test objects do not imply the conclusion (Example 3.19).

Example 3.19. Let Z be the figure-eight curve built, as the earrings above, from the line of slope $1/3$ through $(0, 0)$. This line passes through (π, π) , whereas the line of $Q_{1/3}$ joins $(\pi, 0)$ to $(0, \pi)$, so Z is not the earring $E_{-1/3}$. The objects $(N, b_{S_{18}})$ and $(N, b_{S_{18}} + b_{S_{35}})$ have equal pairing dimensions, 9 and 31, with $\mathcal{E}$ and $\mathcal{E}_*$; the second is one of the 41 two-switch sums of Proposition 3.12 with the pair $(9, 31)$. With Z , however, their pairing dimensions are 9 and 11. Since $\dim H \operatorname{Mor}(Z, -)$ is a homotopy invariant, the two objects are not homotopy equivalent.

By Proposition 3.4, Remark 3.18 becomes usable only after a wrapped representability theorem for immersed correspondences that covers the triple point, or an embedded correspondence inducing the same quilted module; neither is currently available. The full $\mathcal{B}$ -module comparison would then remain, and so would the identification of the symplectic assignment with the instanton tangle invariant, which Cazassus, Herald, Kirk and Kotelskiy state as [CHKK22, Conj. D].

4. COMPARISON WITH THE BAR-NATAN INVARIANT

This section compares the deformed models (N_q, b) of Smith's curve with the Bar-Natan invariant of $T_q = Q_{1/3} + Q_{1/q}$. Throughout, q is odd, $K_r(q) = \operatorname{num}(Q_r + T_q)$ for $r \in \mathbb{Q} \cup \{\infty\}$, so that $K_{-1/2}(q) = P(-2, 3, q)$ and $K_{-3/4}(7) = K_*$, and HF and pairing dimensions are as in §2.3.

4.1. Khovanov invariants in Smith's coordinates. Let G be the group of isometries of $\mathbb{R}^2$ of §2.1, generated by the translations by $2\pi\mathbb{Z}^2$ and the reflection $(\gamma, \theta) \mapsto (-\gamma, -\theta)$, so that the pillowcase is $\mathbb{R}^2/G$ in the coordinates (γ, θ) of §2.2. The group G acts freely on the complement of the *lattice* $\pi\mathbb{Z}^2$, and the quotient map $\mathbb{R}^2 \setminus \pi\mathbb{Z}^2 \rightarrow P^*$ is a covering, which we call the *planar cover*. The lattice point $(m\pi, n\pi)$ lies over the corner whose coordinates are the residues of $m\pi$ and $n\pi$ modulo 2π . A *lattice line* is a line in $\mathbb{R}^2$ through two lattice points. The lattice lines parallel to a given one form two G -orbits: the lines of one orbit pass over two of the corners, and those of the other over the remaining two. Every straight arc between two corners, such as the line of a rational tangle (§2.2), is the image of a segment joining consecutive lattice points of a lattice line.

The translation $\Phi(\gamma, \theta) = (\gamma + \pi, \theta)$ normalizes G and therefore descends to an involution of the pillowcase. It is a symplectomorphism of P^* , it exchanges the corners $(0, 0) \leftrightarrow (\pi, 0)$ and $(\pi, \pi) \leftrightarrow (0, \pi)$, and it is the translation of Remark 3.2.

The Bar-Natan invariant $\widetilde{\operatorname{BN}}(T)$ and the Khovanov invariant $\widetilde{Kh}(T)$ of §2.3 are defined for pointed four-ended tangles T [KWZ19, Def. 6.5]. Both are objects of $\operatorname{Tw}(\mathcal{B})$, and their arc components end at the three non-special punctures. Under the identification of [KWZ19, §8.2], the special puncture corresponds to the corner $(0, \pi)$ in the coordinates of [CHKK22], which is the corner that holonomy-perturbed tangle character varieties miss [CHKK22, Prop. 11.1 and §11.6]. In Smith's coordinates that corner is (π, π) (§2.2). We therefore draw an invariant Y of [KWZ19] in Smith's coordinates as $\Phi(Y)$, and we set

$$\operatorname{BN}_q := \Phi(\widetilde{\operatorname{BN}}(T_q)).$$

For a rational tangle Q , the arc $\widetilde{\operatorname{BN}}(Q)$ and the figure-eight curve $\widetilde{Kh}(Q)$ around it agree, under this identification, with the holonomy-perturbed traceless character varieties of Q and of Q with an earring [KWZ19, §1.8; for $\widetilde{\operatorname{BN}}$ also §8.2, Prop. 8.3]. When Q_r is glued to T_q , its mirror Q_{-r} enters the pairing theorem [KWZ19, Thm. 1.9], and in Smith's coordinates the curve $\Phi(\widetilde{Kh}(Q_{-r}))$ is the earring curve E_r of $\widehat{Q}_r$ used in Section 3, a Herald–Kirk figure-eight [HK24, Def. 6.7].

This identification is cited from [KWZ19, §1.8; for $\widetilde{\text{BN}}$ also §8.2, Prop. 8.3], not proved here. Appendix A.4 records direct comparisons of encodings at finitely many slopes that confirm it and (4.1) below. We call the straight arc that E_r surrounds the *arc of E_r* . For $r = m/n$ in lowest terms it joins $(\pi, 0)$ to $(0, \pi)$ if m and n are odd, $(0, 0)$ to $(0, \pi)$ if n is even, and $(0, 0)$ to $(\pi, 0)$ if m is even (§2.2). The complexes $\mathcal{E} = \mathcal{E}_{-1/2}$ and $\mathcal{E}_* = \mathcal{E}_{-3/4}$ of Section 3 encode $E_{-1/2}$ and $E_{-3/4}$. Since Φ is a symplectomorphism, the pairing theorem therefore reads

$$\dim_{\mathbb{F}_2} HF(E_r, \text{BN}_q) = \text{rank } \widetilde{Kh}(K_r(q); \mathbb{F}_2), \quad r \in \mathbb{Q} \cup \{\infty\}. \quad (4.1)$$

Gradings of curves in the four-punctured sphere depend on which puncture is special. Accordingly, we call a curve Γ in Smith's coordinates *bigraded* if $\Phi^{-1}(\Gamma)$ is bigraded in the sense of [KWZ19]; this is the normalization in which (π, π) is the special puncture.

4.2. The structure theorem. Put $r_q = (1 - q)/(3q - 4)$. Its numerator is even and its denominator is odd, so the arc of E_{r_q} joins $(0, 0)$ to $(\pi, 0)$. We use three curves.

- α_q is the arc of E_{r_q} , an embedded arc from $(0, 0)$ to $(\pi, 0)$.
- c is the boundary of a regular neighborhood of the arc of E_{-1} , which joins $(\pi, 0)$ to $(0, \pi)$. Thus c is a simple closed curve separating $\{(\pi, 0), (0, \pi)\}$ from $\{(0, 0), (\pi, \pi)\}$, and its lifts are lines parallel to the lifts of the arc of E_{-1} that pass through no lattice point.
- (c, J_2) is c with the indecomposable two-dimensional local system $J_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. As a twisted complex it is the *doubled word* of c , the word of c read twice (§2.3); the doubled word represents c with the transposition local system, which over $\mathbb{F}_2$ is similar to J_2 [KWZ19, §5.3], [AS21, Lem. 2.24].

Up to isotopy, Φ carries each of these curves to itself. The arc α_q and the arc BN_q are bigraded, as are all invariants of [KWZ19]. The curve c admits a δ -grading in the sense of [Zib20, Def. 4.28] but no bigrading, and neither does (c, J_2) . Indecomposable two-dimensional local systems also occur on components of $\widetilde{\text{BN}}$ of other tangles, over other fields [KWZ26, Rem. 3.20].

Write N_q for the reduced twisted complex that encodes $L_{q, \text{PL}}$ in the chart of §3.3.2, so that $N_7 = N$; the complex N_5 is displayed in (A.1). Each is an arc from $(0, 0)$ to $(\pi, 0)$, and its end at $(0, 0)$ is the generator 36 of N_7 , respectively 0 of N_5 . In that chart the face R contains the corner $(0, 0)$ (Figure 1), so an arrow labelled R is a chord around the corner $(0, 0)$. The four-arrow element $b_5 \in \text{End}(N_5)$ of (A.2) satisfies $D_{N_5}(b_5) = b_5^2 = 0$, and (N_5, b_5) is strictly isomorphic to the encoding of the oriented smoothing of $L_{5, \text{PL}}$ at each of two of its self-intersection orbits; it raises the pairing dimension with $E_{-1/2}$ from 5 to $7 = q + 2$.

To describe BN_q , fix a lift p_0 of $(0, 0)$ and let p_M be the lift of $(\pi, 0)$ for which the segment $[p_M, p_0]$ is a lift of α_q . Let ℓ be the lattice line through p_0 parallel to the lifts of c ; its lattice points lie alternately over $(0, 0)$ and (π, π) . Number them p_j , $j \in \mathbb{Z}$, consecutively and so that the angle between $p_1 - p_0$ and $p_0 - p_M$ is acute, and let ℓ^+ be the open half-plane bounded by ℓ that does not contain p_M . The translation by $p_2 - p_0 \in 2\pi\mathbb{Z}^2$ generates the stabilizer in G of each lift of c .

Theorem 4.1 (Theorem 1.1). (a) Let $q = 7$ and $b \in \{b_{S_{18}}, b_{S_{25}}\}$, or $q = 5$ and $b = b_5$. Then (N_q, b) is strictly isomorphic to $\alpha_q \oplus (c, J_2)$.

(b) Let $5 \leq q \leq 21$ be odd. Then BN_q is the arc whose lift starts at p_M , runs along $[p_M, p_0]$, crosses ℓ between p_{-1} and p_0 , and then runs in ℓ^+ , parallel to ℓ and close to it, until it ends at p_4 .

(c) Let q and b be as in (a), and let

$$\kappa_7 = 0 \xrightarrow{R} 19 + 36 \xrightarrow{R} 19 \in \text{End}(N_7), \quad \kappa_5 = 12 \xrightarrow{R} 11 + 0 \xrightarrow{R} 11 \in \text{End}(N_5).$$

Then $(\delta_{N_q} + \kappa_q)^2 = (\delta_{N_q} + b + \kappa_q)^2 = 0$, and $(N_q, b + \kappa_q)$ is strictly isomorphic to BN_q .

Part (b) says that BN_q arises from α_q by sliding its end at $(0, 0)$ twice around that corner, parallel to c : beyond p_0 the lift covers two periods of a lift of c , ending at the lift $p_4 = p_0 + 2(p_2 - p_0)$ of $(0, 0)$. In (c), the arrows $0 \xrightarrow{R} 19$ and $12 \xrightarrow{R} 11$ are terms of δ_{N_7} and δ_{N_5} , so adding κ_q deletes one arrow and adds an arrow labelled R at the end of N_q at $(0, 0)$. Both terms of κ_q are chords around the corner $(0, 0)$.

Proof. Each statement asserts that two explicit reduced twisted complexes are strictly isomorphic, or that an explicit matrix over $\mathcal{B}$ squares to zero. The complexes of α_q , of c , of its doubled word and of the arc in (b) are obtained by encoding their lifts in the chart of §3.3.2; $\widetilde{\text{BN}}(T_q)$ is computed with Zibrowius's program [Zib] and drawn in Smith's coordinates by Φ . All complexes involved are of loop type (§2.3), so a strict isomorphism is decided by comparing the cyclic or linear words of the components, matching idempotents, labels and the directions of arrows. For $q = 5$ the complexes in (a) and (c) are printed in Appendix A.1, where these comparisons can be checked by hand.

For (a), $(N_7, b_{S_{18}})$ and $(N_7, b_{S_{25}})$ have components of 23 and 8 generators, and (N_5, b_5) of 15 and 8; in each case the first component is strictly isomorphic to α_q and the second to the doubled word of c , which represents (c, J_2) . For (b), the encoding of the arc described there is strictly isomorphic to BN_q for $q = 5, 7, \dots, 21$; it is the only one among the 24 arcs obtained from α_q by sliding one of its ends along c , in either direction and on either side, through one, two or three periods. For (c), direct multiplication in $\mathcal{B}$ gives the two Maurer–Cartan identities, and the words of $(N_q, b + \kappa_q)$ and of BN_q agree. $\square$

Remark 4.2 (The corner term). The arrow that κ_q deletes is a chord of the closed component of (N_q, b) around the corner $(0, 0)$. Deleting it opens that component into an arc with two ends at $(0, 0)$, and the added arrow joins one of them to the end of α_q . Thus κ_q changes which two of the three arc ends at $(0, 0)$ are joined. An exhaustive search over the complexes obtained from (N_q, b) by deleting one arrow and adding one arrow between two generators that meet at most one arrow finds, for each of $b_{S_{18}}$, $b_{S_{25}}$ and b_5 , exactly two that are strictly isomorphic to BN_q (566 candidates for each b at $q = 7$, and 402 at $q = 5$). In each, the added arrow is labelled R and starts at the end of N_q at $(0, 0)$; κ_7 is the one common to $b_{S_{18}}$ and $b_{S_{25}}$, and the others are listed in §A.2.

Of the 45 distinct switches of oriented smoothings at $q = 7$, 41 are algebraic switches (§3.3.3); each of the other four exchanges the heads of an arrow labelled M and an arrow labelled M^2 . The term κ_q is not a sum of switches: no sum of at most three algebraic switches of N_q is strictly isomorphic to BN_q (there are 110 algebraic switches of N_7 and 221 925 such sums, and 58 switches of N_5 and 32 567 sums), and no sum of at most three of these 45 switches is strictly isomorphic to BN_7 .

Remark 4.3 (The opposite sign of the perturbation). By Remark 3.17, the pairing dimensions for small $t > 0$ are those of the curves $\Phi(E_r)$ with the present objects. Since Φ is an involution, $\Phi(E_r) = \widetilde{Kh}(Q_{-r})$, and the comparison object becomes $\Phi(\text{BN}_q) = \widetilde{\text{BN}}(T_q)$. The curves α_q and c are Φ -invariant, so part (a) is unchanged; in (b) the slide takes place at the end $(\pi, 0)$; and (c) holds with

$$\kappa'_7 = 2 \xrightarrow{M} 3 + 18 \xrightarrow{M^2} 3, \quad \kappa'_5 = 10 \xrightarrow{M} 11 + 22 \xrightarrow{M^2} 11$$

in place of κ_7 and κ_5 , where 18 and 22 are the ends at $(\pi, 0)$. The two corner terms are not interchangeable. For $t < 0$, the object $(N_7, b_{S_{18}} + \kappa'_7)$ has pairing dimension 13 with E_{-1} , whereas $\text{rank } \widetilde{Kh}(K_{-1}(7)) = 11$; under Conjecture D of [CHKK22] this pairing dimension would be $\dim I^\natural(K_{-1}(7))$, contradicting $\dim I^\natural \leq \text{rank } \widetilde{Kh}$ [KM11b]. At $q = 5$ the corresponding values are 9 and 7.

4.3. Pairing with linear curves. For curves X, Y in P^* , at least one of them compact, let $i(X, Y)$ be the minimal number of intersection points of curves homotopic to X and Y , where arcs are homotoped through arcs with the same ends.

Definition 4.4. Let $w \in \mathbb{Z}^2$ be primitive. A compact curve $\Gamma \subset P^*$ is *linear of direction w* if every lift of Γ is a graph over a line of direction w and lies in an open strip about that line that contains no lattice point off the line, and either the line meets no lattice point, in which case Γ is the simple closed curve of direction w , or the line is a lattice line and the lift passes each of its lattice points on a prescribed side, the choices of side being invariant under G .

Every simple closed curve is linear. So is every earring E_r , whose lifts pass the consecutive lattice points of the lattice line containing the lifts of its arc on alternating sides, and more generally every figure-eight around a straight arc between two corners.

Theorem 4.5 (Theorem 1.2). *Let $5 \leq q \leq 21$ be odd, and let Γ be a linear curve whose direction is parallel neither to α_q nor to c .*

- (a) $i(\Gamma, \text{BN}_q) = i(\Gamma, \alpha_q) + 2i(\Gamma, c)$.
- (b) *If Γ is bigraded, that is, if $\Phi^{-1}(\Gamma)$ is bigraded in the sense of [KWZ19], then, granting the arc case of the intersection formula of Kotelskiy, Watson and Zibrowius [KWZ19, Thm. 5.25],*

$$\dim HF(\Gamma, \text{BN}_q) = \dim HF(\Gamma, \alpha_q) + 2 \dim HF(\Gamma, c) = \dim HF(\Gamma, \alpha_q \oplus (c, J_2)).$$

Proof. Part (b) from part (a). For two δ -graded compact curves that are not parallel, with local systems X and X' , the dimension of HF is $(\dim X)(\dim X')$ times their minimal intersection number [Zib20, Thm. 4.45]; this applies to Γ and (c, J_2) and gives $\dim HF(\Gamma, (c, J_2)) = 2i(\Gamma, c) = 2 \dim HF(\Gamma, c)$. For bigraded curves one of which is an arc, the same formula is [KWZ19, Thm. 5.25], applied after moving all curves back by Φ ; Kotelskiy, Watson and Zibrowius state it under the standing assumption that both curves are compact, adding that the other cases are treated similarly, and prove it by referring to the proof of [Zib20, Thm. 4.45], which is set on a marked surface whose paths end on the boundary rather than at punctures; neither paper writes out the case of an arc ending at a puncture, which is the arc case granted in (b). It applies to Γ and the bigraded arcs α_q and BN_q . Hence (a) implies (b).

Geodesic representatives. Fix a small $\varepsilon > 0$ and remove from $\mathbb{R}^2$ the open ε -discs $B(p)$ about the lattice points p . Their boundary circles are locally concave, so the flat surface that remains is locally CAT(0) by the theorem of Alexander, Berg and Bishop [AKP26, Thm. 7.28], and its universal cover is CAT(0) by the globalization theorem [AKP26, Thm. 5.6]. Two geodesics in a CAT(0) space meet in a connected set, and two geodesic lifts without a common boundary arc meet if and only if their ends link. Hence geodesic representatives in the quotient by G , in which an arc ends orthogonally on the boundary circle of its corner, realize the minimal intersection numbers, provided a common arc of tangency along a boundary circle is counted once if the four ends of the two curves near it interleave and not at all otherwise. The representatives are taut curves made of straight segments and short arcs of the boundary circles: for Γ , a line of direction w missing the discs if Γ is the simple closed curve, and otherwise the curve that passes each disc of its lattice line on the prescribed side, which is a line of direction w tangent to the discs when all sides agree and in general consists of common tangent segments of consecutive discs and short arcs of their boundary circles; for α_q , the part of $[p_M, p_0]$ outside the two discs; for c , the line in ℓ^+ parallel to ℓ at distance ε ; and for BN_q , the segment from $\partial B(p_M)$ to a tangent point of $\partial B(p_0)$, an arc of $\partial B(p_0)$, the same parallel line, and near p_4 a short arc of $\partial B(p_3)$ followed by a segment that ends orthogonally on $\partial B(p_4)$, arriving nearly along the direction of $p_4 - p_3$. The segment $[p_M, p_0]$ meets no other disc because its direction is primitive.

Reduction to one lattice point. A lift of an arc has trivial stabilizer, so for an arc X the number $i(\Gamma, X)$ counts the intersections of the full preimage of Γ with one lift of X . A fundamental domain for the stabilizer of the lift of c is one period, so $2i(\Gamma, c)$ counts the intersections with the half-open stretch of the parallel line from p_0 to p_4 , the end near p_0 included. Compare the lift of BN_q with the union of the lift of α_q and this stretch. Outside small neighborhoods of $p_M, p_0, \dots, p_4$ the two coincide. Near p_M both end along the same segment, near p_1, p_2 both follow the parallel line, and near p_3, p_4 the lift of BN_q turns along $\partial B(p_3)$ to end on $\partial B(p_4)$. The translation by $p_4 - p_0 \in 2\pi\mathbb{Z}^2$ belongs to G , so it carries the lifts of Γ through p_4 bijectively onto those through p_0 , preserving the sides. It therefore suffices to show, for every lift λ of Γ through p_0 , that

$$\text{corner}(u, v) + \text{end}(-v) = \text{end}(u) + 1, \quad (4.2)$$

where u and v are the directions from p_0 to p_M and to p_1 ; $\text{end}(x)$ is the number of intersections of λ with an arc that ends on $\partial B(p_0)$ along the direction x ; $\text{corner}(u, v)$ is the number of intersections of λ with the part of BN_q near p_0 ; and the summand 1 counts the straight passage of the stretch of c .

Local counts. Let D be the open half-plane bounded by the lattice line of λ into which λ is displaced near p_0 , and D' the other one. The directions u and v are not parallel to w , so each points into D or into D' . Near p_0 the curve λ is a graph over its lattice line with slope $O(\varepsilon)$, and it touches or wraps around $\partial B(p_0)$ on the side of D . An arc ending orthogonally on $\partial B(p_0)$ along x meets it once if x points into D , at its endpoint when x is orthogonal to w , and not at all otherwise. A straight passage, which crosses the lattice line of λ at an angle bounded below, meets it once. The angle between u and v is less than π , and the part of BN_q near p_0 runs in along u , around the far side of $B(p_0)$ from that angle, and out along v ; it meets λ once if u and v point into different half-planes, twice if both point into D , and not at all if both point into D' . Since $\text{end}(-v) = 1$ exactly when v points into D' , the two sides of (4.2) are

	$u \in D, v \in D'$	$u \in D', v \in D$	$u, v \in D$	$u, v \in D'$
left side	$1 + 1$	$1 + 0$	$2 + 0$	$0 + 1$
right side	$1 + 1$	$0 + 1$	$1 + 1$	$0 + 1$

and they agree in every case. This proves (a). $\square$

Corollary 4.6 (Corollary 1.3). *Let $q \in \{5, 7\}$ and let b be as in Theorem 4.1(a). Granting the arc case of the intersection formula of Kotelskiy, Watson and Zibrowius [KWZ19, Thm. 5.25], for every $r \in \mathbb{Q} \cup \{\infty\}$,*

$$\dim HF(E_r, (N_q, b)) = \dim HF(E_r, (N_q, b + \kappa_q)) = \text{rank } \widetilde{Kh}(K_r(q); \mathbb{F}_2).$$

Proof. The second equality is Theorem 4.1(c) together with (4.1). The earring E_r is linear, and after moving all curves back by Φ it becomes the bigraded curve $\widetilde{Kh}(Q_{-r})$. If its direction is parallel neither to α_q nor to c , Theorems 4.1(a) and 4.5(b) give $\dim HF(E_r, (N_q, b)) = \dim HF(E_r, \text{BN}_q)$. The two remaining earrings are E_{-1} , parallel to c , and E_{r_q} , parallel to α_q , and for them all three numbers are computed directly:

q	E_{-1}			E_{r_q}		
	(N_q, b)	$(N_q, b + \kappa_q)$	$\text{rank } \widetilde{Kh}$	(N_q, b)	$(N_q, b + \kappa_q)$	$\text{rank } \widetilde{Kh}$
5	7	7	7	30	30	30
7	11	11	11	46	46	46

The undeformed complex N_q has pairing dimensions 7 and 28 with E_{-1} and E_{r_q} at $q = 5$, and 11 and 44 at $q = 7$. $\square$

We have also computed the three numbers directly at 340 slopes for $q = 5$ and 342 slopes for $q = 7$: every slope with $|r| \leq 2$ and denominator at most 14, every slope with $|r| \leq 5$ and denominator at most 6, the integers with $|r| \leq 10$, $\pm r_q$ and ∞ . These include 108 slopes with odd numerator and odd denominator, for which the arc of E_r joins $(\pi, 0)$ to $(0, \pi)$. The three numbers agree at every slope. The pairing dimension of E_r with the undeformed complex N_q is less than $\text{rank } \widetilde{Kh}$ at 40 of these slopes for $q = 5$ and at 43 for $q = 7$, and never more. Under Conjecture D of [CHKK22] the pairing dimension of E_r with the assigned object is $\dim I^\natural(K_r(q))$, which is at most $\text{rank } \widetilde{Kh}(K_r(q))$ [KM11b]. Both (N_q, b) and $(N_q, b + \kappa_q)$ attain this bound at every computed slope and, granting the arc case of the intersection formula [KWZ19, Thm. 5.25], at every rational closure (Corollary 4.6): at $r = -1/2$ the common value is $q + 2 = \dim I^\natural(P(-2, 3, q))$ [Wue26, Thm. 1.1], and at $r = -3/4$ and $q = 7$ it is $31 = \dim I^\natural(K_*)$ (Proposition 3.8).

Remark 4.7 (The direction of c). The hypothesis that the direction is not parallel to c cannot be dropped. Among the figure-eights around straight arcs and the simple closed curves in the enumeration described below (directions (m, n) with $|m|, |n| \leq 6$ or parallel to α_q , and $q = 5, 7, \dots, 13$), exactly two curves have different pairing dimensions with BN_q and with $\alpha_q \oplus (c, J_2)$, and both are parallel to c : the curve c itself, with 11 against 13 at $q = 7$ and 7 against 9 at $q = 5$, and the figure-eight around the arc parallel to c that joins $(0, 0)$ to (π, π) , with 13 against 11 and 9 against 7. The latter surrounds the corner (π, π) , which no tangle character variety meets, and is not the earring of a rational tangle; the earring E_{-1} , around the arc joining $(\pi, 0)$ to $(0, \pi)$, has equal pairing dimensions. For general *side patterns*, the G -invariant choices of sides in Definition 4.4 read along the lattice line, we have computed all linear curves with direction (m, n) , $|m|, |n| \leq 6$, or parallel to α_q , on lattice lines of both orbits and with every side pattern of period 2, 4 or 6, together with the simple closed curves: 1887 curves for each $q = 5, 7, \dots, 13$. For each q , exactly 22 of them have different pairing dimensions, all parallel to c : all 19 side patterns on the lattice line through $(0, 0)$ and (π, π) , the two constant patterns on the line through $(\pi, 0)$ and $(0, \pi)$, and the simple closed curve. The constant patterns on either line and the simple closed curve are isotopic to c . No curve parallel to α_q has different pairing dimensions, although the proof does not cover that direction.

Remark 4.8 (Curves parallel to c). Curves parallel to c separate (N_q, b) from $(N_q, b + \kappa_q) \cong BN_q$. At $q = 7$ the curve c has pairing dimension 11 with BN_7 and 13 with $(N_7, b_{S_{18}})$. For (c, J_2) the pairing dimensions are 22 and 26, the difference being $\dim HF((c, J_2), (c, J_2)) = 4$. The image under Φ of the special component of $\widetilde{Kh}(T_7)$ in the sense of [KWZ26, §3.4, Def. 3.11], the component other than the figure-eight $\widetilde{Kh}(Q_{-r_7})$, is parallel to c and has pairing dimensions 52 and 44. At $q = 5$ the corresponding values are 7, 9; 14, 18; and 36, 28.

4.4. The object assigned to T_q . Theorem 4.1 and Corollary 4.6 suggest, without deciding, the following question.

Question 4.9 (Question 1.4). Let $q \geq 5$ be odd with $\gcd(q, 3) = 1$. Suppose that the tangle assignment of Cazassus, Herald, Kirk and Kotelskiy is defined for T_q with Smith's perturbation for small $t < 0$, and let X_q be the object of $\text{Tw}(\mathcal{B})$ that it determines, an immersed curve with a bounding cochain, drawn in Smith's coordinates. Is X_q homotopy equivalent to BN_q ?

For $q = 5$ and $q = 7$ the evidence is as follows. The object BN_q arises from the piecewise-linear model of Smith's curve by the explicit Maurer–Cartan element $b + \kappa_q$ (Theorem 4.1). By Corollary 4.6, which grants the arc case of the intersection formula [KWZ19, Thm. 5.25], (N_q, b)

and $(N_q, b + \kappa_q)$ have the same pairing dimension with every rational earring, equal to the rank of reduced Khovanov homology of the closure; both are compatible with $\dim I^\natural \leq \text{rank } \widetilde{Kh}$ at every rational closure, and rational closures cannot distinguish them. They are distinguished by curves parallel to c (Remark 4.8), and by gradings: BN_q is bigraded and (N_q, b) is not (and at $q = 7$ the only bigraded deformation among the four of Proposition 3.7, the S_{69} class, fails the second closure), but no quantum grading is known on the instanton side, so this difference is not a test.

Proposition 4.10. *The object BN_7 is not homotopy equivalent to any object allowed by Hypothesis 3.13. Granting the arc case of the intersection formula of Kotelskiy, Watson and Zibrowius [KWZ19, Thm. 5.25], $\dim HF(E_r, BN_7) = \dim HF(E_r, (N, b_{S_{18}}))$ for every $r \in \mathbb{Q} \cup \{\infty\}$.*

Proof. The dimension of $HF(Z, -)$ for a fixed object Z is invariant under homotopy equivalence. By (4.1), the pairing dimensions of $E_{-1/2}$ and $E_{-3/4}$ with BN_7 are 9 and 31, the ranks of reduced Khovanov homology of $P(-2, 3, 7)$ and K_* ([Wue26, §3] and Proposition 3.8). Among the objects allowed by Hypothesis 3.13, only $(N, b_{S_{18}})$ and $(N, b_{S_{25}})$ have this pair (Propositions 3.10–3.12, as in the proof of Corollary 3.14), and they are strictly isomorphic (Corollary 3.9). By Theorem 4.1(a),

$$\dim HF((c, J_2), (N, b_{S_{18}})) = \dim HF((c, J_2), \alpha_7) + \dim HF((c, J_2), (c, J_2)) = 22 + 4 = 26,$$

whereas $\dim HF((c, J_2), BN_7) = 22$. The last statement is Corollary 4.6. $\square$

Consequently an affirmative answer to Question 4.9 for $q = 7$ would show that Hypothesis 3.13 fails. The two closures of Corollary 3.14 assign the same dimensions to BN_7 as to the class of $(N, b_{S_{18}})$ that the hypothesis selects, and, granting the arc case of the intersection formula [KWZ19, Thm. 5.25], so does every other rational closure; no rational closure could then detect the failure.

Remark 4.11 (The cases $q \geq 11$). For $q = 11$ and 13 the encoding of a numerical trace of Smith’s curve is strictly isomorphic to N_q , with 47 and 55 generators, and its pairing dimension with $E_{-1/2}$ is $q + 4$, two more than $\text{rank } \widetilde{Kh}(P(-2, 3, q)) = q + 2$; for $q = 17$ and 19 the pairing dimension of the traced curve is $q + 8$. At $q = 17, 19$ we have not computed the encoding of $L_{q, \text{PL}}$, and we write N_q^{tr} for the encoding of the traced curve; the comparison of traces with N_q in Appendix A covers $q \leq 13$. At $q = 5, 7$ the pairing dimension is q , and one switch raises it to $q + 2$. The deformation of Theorem 4.1 does not persist in the following finite sense. Call the minimal number of arrows that must be deleted from δ_{N_q} , and the same number added, to reach a complex strictly isomorphic to BN_q the *edit distance*. It is 2 for $q = 5, 7$, 4 for $q = 11, 13$, and at least 6 for $q = 17, 19$, where N_q^{tr} takes the place of N_q . At $q = 5, 7$ one algebraic switch followed by one replacement of an arrow reaches BN_q ; at $q = 11$ and 13 no chain of at most three algebraic switches, each Maurer–Cartan for the complex it deforms, ends one replacement away from BN_q . At $q = 11$ the first such chains have four switches, and they end at the sum of an arc and a closed curve other than $\alpha_{11} \oplus (c, J_2)$; the latter is more than five switches away from N_{11} , against one at $q = 5, 7$. None of the 35 004 complexes (6 331 up to strict isomorphism) obtained from N_{11} by at most two algebraic switches has α_{11} as a summand, and for each of them either the pairing dimension with $E_{-1/2}$ differs from $\dim I^\natural(P(-2, 3, 11)) = 13$, or the pairing dimension with E_r exceeds $\text{rank } \widetilde{Kh}(K_r(11))$ for some r in a fixed set of 25 slopes. These are finite searches within the class of algebraic switches, and they do not exclude a larger bounding cochain.

5. DISCUSSION

5.1. Relation to earlier work. For the decomposition of [Wue26, §4], obtained from Hedden, Herald and Kirk’s decomposition of $T(3, 5)$ by regluing, the pillowcase Floer homology with zero

bounding cochain already agrees with $I^\natural(K_q)$ in every grading, for odd $q \geq 7$ [Wue26, Thm. 1.2] and for $q = 5$ by the computation of Hedden, Herald and Kirk [HHK18, §11]; no bounding cochain is needed there.

Smith proves that for $q = 5$ his decomposition $\widehat{Q_{-1/2}} \cup (Q_{1/3} + Q_{1/5})$ with zero bounding cochains gives a Floer homology of rank 5 rather than $\dim I^\natural = 7$ [Smi24, proof of Thm. 5.9], so that under Conjecture D of [CHKK22] a nonzero bounding cochain is required [Smi24, Thm. 5.9]; in his closing paragraph he proposes finding the bounding cochains that correct the rank. Sections 3 and 4 give two answers at $q = 5$ and $q = 7$ that, granting the arc case of the intersection formula [KWZ19, Thm. 5.25], no rational closure distinguishes: the smoothings (N_q, b) , and their deformations $(N_q, b + \kappa_q) \cong \text{BN}_q$.

For rational tangles the traceless character varieties coincide with the invariants of Kotelskiy, Watson and Zibrowius [KWZ19, §1.8]. Hedden, Herald, Hogancamp and Kirk assemble Khovanov’s tangle invariant from the traceless character varieties of the crossingless resolutions [HHHK20, Thm. 1.1 and Cor. 1.2], with a result that agrees with $\widetilde{Kh}(T)$ [KWZ20], and they ask whether higher terms can be added so as to recover the Kronheimer–Mrowka spectral sequence [HHHK20, Question 1.4]. Cazassus, Herald, Kirk and Kotelskiy conjecture curve-valued instanton invariants $I(T)$ and $I^\natural(T)$ of tangles, and predict that the relevant algebra is that of [KWZ19] [CHKK22, Conj. E and §11.4]. They do not conjecture $I(T) = \widetilde{\text{BN}}(T)$, which cannot hold for all tangles because $\dim I^\natural < \text{rank } \widetilde{Kh}$ for some knots, for instance $T(4, 5)$ [CHKK22, §§10.2–10.3]. Question 1.4 concerns one tangle, T_q , for whose closures $P(-2, 3, q)$ and K_* the two dimensions agree ([Wue26, §3] and Proposition 3.8). Kotelskiy, Watson and Zibrowius suggest that the differentials of the spectral sequences from Khovanov homology to Heegaard Floer and instanton homology might be realized by smoothing self-intersections of $\widetilde{\text{BN}}(T)$ and $\widetilde{Kh}(T)$ [KWZ19, §1.8]. Sections 3 and 4 proceed in the opposite direction: a smoothing of the curve on the instanton side reaches the Khovanov rank, and one further term carries it to the Bar-Natan invariant BN_q .

Smoothing a self-intersection of an immersed curve by a degree-one bounding cochain is Lagrangian surgery. Palmer and Woodward prove that immersed Floer cohomology is invariant under surgery for Lagrangians of dimension at least three, and in dimension two under an additional assumption [PW26, Thm. 1.3]; for curves on surfaces, Auroux and Smith compare surgery with the mapping cone, which involves a bounding cochain [AS21, Lem. 2.25 and Rem. 2.26]. The corner term κ_q is of a different kind: it is supported on chords around a corner (Remark 4.2), which are morphisms of the wrapped category of [CHKK22, §11.5] and not self-intersection points of the curve.

5.2. Open problems. *The assigned object.* Granting the arc case of the intersection formula of Kotelskiy, Watson and Zibrowius [KWZ19, Thm. 5.25], Question 1.4 cannot be decided by rational closures at $q = 5, 7$ (Corollary 4.6); curves parallel to c , and gradings, separate the candidates (Remark 4.8). For $q \geq 11$ a deformation of Smith’s curve to BN_q , if it exists, is not small in the sense of Remark 4.11.

The support of the bounding cochain. Proposition 4.10 shows that Hypothesis 3.13 is not implied by closure data. A direct proof would have to identify the rigid figure-eight boundary strata of the quilted moduli spaces, exclude simultaneous support at several self-intersection orbits, and intertwine the output with the switches of §3.3.3. In the strip-shrinking model the zero-width boundary stratum is predicted to contain quilted trajectories with any positive number of figure-eight bubbles, and the zero-input correction a count of figure-eight bubble trees [BW18, §§4.2–4.4], so neither zero input nor index zero forces support at a single self-intersection orbit. A proof must use the geometry near the six internal edge circles.

The formal route. Remark 3.18 reduces the identification of the assigned object to a wrapped representability theorem of the kind Gao proves [Gao17, Thm. 7.24 and Cor. 7.25], and Gao’s

theorem requires an embedded correspondence; by Proposition 3.4 the correspondence induced by the line of $Q_{1/3}$ has a triple point. An immersed version of Gao's theorem covering this triple point, or an embedded replacement inducing the same quilted module, would be needed, and neither is available. Independently, every selection statement of Sections 3 and 4 is conditional on the correspondence between instanton and pillowcase invariants [CHKK22, Conj. D]; neither Fukaya's nor Gao's correspondence theorem [Fuk25, Gao17] compares the two.

Gradings of bounding cochains. In the normalization of Section 4 the object $(N, b_{S_{69}})$ is bigraded, and the class of $b_{S_{69}}$ spans the quantum-degree-zero part of the δ -degree -1 endomorphism cohomology of N . The S_{18}/S_{25} and S_{74} objects are not bigraded (§3.3.3, Remark 3.16).

A selection by homogeneity of degree zero for the grading of N would therefore choose the S_{69} class, which the second closure excludes (Proposition 3.8). Bigradedness of the deformed object alone does not select: among the objects allowed by Hypothesis 3.13 with pairing dimension nine with $\mathcal{E}$, the bigraded ones also include $(N, b_{S_{18}} + b_{S_{74}})$ and the switches at S_{34} , S_{38} , S_{75} and S_{77} , all with pairing dimension 25 with $\mathcal{E}_*$. None of the bigraded allowed objects has the pair $(9, 31)$. For $t > 0$, with $(0, \pi)$ special (Remark 4.3), the same holds with S_{10} in place of S_{69} .

No quantum grading is known on the instanton side, so this is an observation and not a test. A $\mathbb{Z}/4$ -graded selection principle compatible with the absolute grading of Hedden, Herald and Kirk [HHK18, §6.1] would replace the closure arguments of Section 3 by a structural one.

APPENDIX A. DATA AND COMPUTATIONS

A.1. The complex N_5 . Encoding $L_{5, \text{PL}}$ in the chart of §3.3.2 and performing the unit cancellations gives the reduced twisted complex N_5 on the generators $0, 1, \dots, 22$. Its $\iota_\circ$ generators are $3, 4, 9, 10, 13, 14, 19, 20$, the others carry $\iota_\bullet$, and

$$\begin{aligned} \delta_{N_5} = & 1 \xrightarrow{M^2} 0 + 2 \xrightarrow{R} 1 + 3 \xrightarrow{M} 2 + 4 \xrightarrow{L} 3 + 4 \xrightarrow{M} 5 + 5 \xrightarrow{R} 6 + 6 \xrightarrow{M^2} 7 + 8 \xrightarrow{R} 7 \\ & + 9 \xrightarrow{M} 8 + 10 \xrightarrow{L} 9 + 10 \xrightarrow{M} 11 + 12 \xrightarrow{R} 11 + 13 \xrightarrow{M} 12 + 14 \xrightarrow{L} 13 + 14 \xrightarrow{M} 15 \\ & + 16 \xrightarrow{R} 15 + 17 \xrightarrow{M^2} 16 + 18 \xrightarrow{R} 17 + 19 \xrightarrow{M} 18 + 20 \xrightarrow{L} 19 + 20 \xrightarrow{M} 21 + 21 \xrightarrow{R} 22. \end{aligned} \tag{A.1}$$

No arrow ends where an arrow with the same face label begins, so $\delta_{N_5}^2 = 0$. The complex is a single arc with ends 0 , at $(0, 0)$, and 22 , at $(\pi, 0)$. The encoding of a numerical trace of Smith's perturbed curve at $q = 5$, computed as in Remark 3.15, is strictly isomorphic to N_5 . Its pairing dimensions with $E_{-1/2}$ and $E_{-3/4}$ are 5 and 15 ; the first agrees with Smith's rank 5 for this decomposition [Smi24, proof of Thm. 5.9].

The smoothing used in Theorem 4.1 is the four-arrow element

$$b_5 = 8 \xrightarrow{R} 7 + 16 \xrightarrow{R} 15 + 8 \xrightarrow{R} 15 + 16 \xrightarrow{R} 7, \tag{A.2}$$

which exchanges the targets of the arrows $8 \xrightarrow{R} 7$ and $16 \xrightarrow{R} 15$. The complex (N_5, b_5) consists of the arc $0, 1, \dots, 7, 16, 17, \dots, 22$ and the closed component $8, 9, \dots, 15$, whose word $9 \xrightarrow{M} 8, 10 \xrightarrow{L} 9, 10 \xrightarrow{M} 11, 12 \xrightarrow{R} 11, 13 \xrightarrow{M} 12, 14 \xrightarrow{L} 13, 14 \xrightarrow{M} 15, 8 \xrightarrow{R} 15$ is the doubled word of c . Adding $\kappa_5 = 12 \xrightarrow{R} 11 + 0 \xrightarrow{R} 11$ opens the closed component between 11 and 12 and attaches 11 to the end 0 ; the result is the single arc $22, 21, \dots, 16, 7, 6, \dots, 0, 11, 10, 9, 8, 15, 14, 13, 12$. Its pairing dimensions with $E_{-1/2}$ and $E_{-3/4}$ are 7 and 21 , the ranks of reduced Khovanov homology of $P(-2, 3, 5)$ and $K_{-3/4}(5)$.

A.2. Corner terms. The search of Remark 4.2 finds the following one-arrow replacements carrying (N_q, b) to a complex strictly isomorphic to BN_q , written as elements of $\text{End}(N_q)$ (the first term is the deleted arrow):

object	first solution	second solution
$(N_7, b_{S_{18}})$	$\kappa_7 = 0 \xrightarrow{R} 19 + 36 \xrightarrow{R} 19$	$22 \xrightarrow{R} 3 + 36 \xrightarrow{R} 3$
$(N_7, b_{S_{25}})$	$\kappa_7 = 0 \xrightarrow{R} 19 + 36 \xrightarrow{R} 19$	$4 \xrightarrow{R} 3 + 36 \xrightarrow{R} 3$
(N_5, b_5)	$\kappa_5 = 12 \xrightarrow{R} 11 + 0 \xrightarrow{R} 11$	$8 \xrightarrow{R} 15 + 0 \xrightarrow{R} 15$

The arrows $22 \xrightarrow{R} 3$ and $8 \xrightarrow{R} 15$ are arrows of $b_{S_{18}}$ and b_5 rather than of δ_{N_q} , so the second solutions in the first and third rows delete an arrow of the smoothing. The term κ_7 is the only one that serves both smoothings at $q = 7$. The analogous search with $\widetilde{\text{BN}}(T_q)$ in place of BN_q , which is the comparison for $t > 0$ (Remark 4.3), finds for both smoothings at $q = 7$ the terms $2 \xrightarrow{M} 3 + 18 \xrightarrow{M^2} 3$ and $20 \xrightarrow{M} 19 + 18 \xrightarrow{M^2} 19$, and at $q = 5$ the terms $10 \xrightarrow{M} 11 + 22 \xrightarrow{M^2} 11$ and $14 \xrightarrow{M} 15 + 22 \xrightarrow{M^2} 15$. The exact edit distance from N_q to BN_q is 2 for $q = 5, 7$, with two minimal edits in each case; at $q = 7$ both consist of arrows labelled R and touch the end 36.

A.3. Finite polygon counts and candidate supports. For $q = 5, 7, 11, 13$ we record the output of a finite combinatorial test on the piecewise-linear curves $R_{\text{PL}} = \mathcal{R}^{\natural}(\widehat{Q_{-1/2}})$ and $L_{q,\text{PL}}$; at $q = 7$ it is superseded by the exact calculation of Proposition 3.7. De Silva, Robbin and Salamon identify holomorphic strips with combinatorial lunes for transverse embedded loops on a surface [dSRS14]; their hypotheses do not cover immersed curves in the orbifold pillowcase, nor the higher polygons of a bounding-cochain deformation. The statements labelled Computation below record outputs of this test as data.

A.3.1. The finite winding test. For intersection points x, y choose an arc α on R_{PL} from x to y and an arc β on $L_{q,\text{PL}}$ from y to x , and lift $\alpha \cdot \beta$ to the torus. The pair is a *candidate bigon* when (a) the lifted loop is null-homologous, (b) its winding function vanishes at the four corners, (c) that function is nonnegative on the sampled complementary regions, and (d) the local sectors at x and y are convex. The same conditions at every vertex define candidate triangles and quadrilaterals; we call their conjunction the *low-order test*. The matrices use one fixed lift of R_{PL} and sum over the two torus preimages of a self-intersection of $L_{q,\text{PL}}$. The arcs α and β range over explicitly listed subarcs of the two curves, each traversed at most once, and no geometric bound shows that these subarcs exhaust all candidates. We write A_q for the set of self-intersections of $L_{q,\text{PL}}$ that are vertices of a candidate triangle or quadrilateral found in this way. The parameters $(\varepsilon_L, \varepsilon_R, r_R)$, namely the connector amplitude of $L_{q,\text{PL}}$ and the perturbation amplitude and pinch parameter of R_{PL} , are $(0.05, 0.10, 0.25)$ at $q = 5$ and $(0.05, 0.16, 0.40)$ at $q = 7, 11, 13$ unless stated otherwise. With C_q the $\mathbb{F}_2$ -vector space on the transverse intersections and D_q^{big} the matrix of candidate bigons, put

$$h_q^{\text{big}} := \dim_{\mathbb{F}_2} C_q - 2 \text{rank}_{\mathbb{F}_2} D_q^{\text{big}}; \quad (\text{A.3})$$

this is a number attached to a finite matrix, not the homology of a Floer complex.

At $q = 5$ the curves meet in nine points (one interior, two clusters of four near $\gamma = \pi$), and the test accepts exactly two candidate bigons with distinct vertices, so $h_5^{\text{big}} = 9 - 2 \cdot 2 = 5$; this agrees with the nine intersections and two bigons in Smith’s proof of [Smi24, Thm. 5.9]. The generator counts depend on the parameters: at $q = 11$ the matrix D_{11}^{big} has 19 generators and rank two at the parameters $(0.05, 0.16, 0.40)$, and 23 generators and rank four at the parameters $(0.03, 0.10, 0.25)$, and at $q = 13$ there are 27 against 25 generators; at $q = 11$ the model has 240 self-intersections, some less than 10^{-3} apart. The value of h_q^{big} agrees at two independent parameter choices, at both of which the matrices D_q^{big} for $q = 5, 7, 11, 13$ square to zero. One

non-generic earring parameter choice at $q = 7$, which collapses the 13 generators to nine, was excluded.

Computation A.1 (Bigon ranks). For the specified representatives and parameters,

$$h_5^{\text{big}} = 5, \quad h_7^{\text{big}} = 7, \quad h_{11}^{\text{big}} = 15, \quad h_{13}^{\text{big}} = 17,$$

and the matrices D_q^{big} square to zero over $\mathbb{F}_2$. The value 5 at $q = 3$ in Table 1 is the computation of Hedden, Herald and Kirk [HHK18, §11.6] from a different decomposition; it is not an output of this family.

q	$\det K_q$	h_q^{big}	$\dim_{\mathbb{F}_2} I^\natural$	difference	required rank change
3	3	5 (external)	5	0	0
5	1	5	7	-2	-1
7	1	7	9	-2	-1
11	5	15	13	+2	+1
13	7	17	15	+2	+1

TABLE 1. Bigon ranks and instanton dimensions. The instanton column is [Wue26, Thm. 1.1]. The column h_q^{big} for $q \geq 5$ concerns the specified finite matrices; the $q = 3$ entry is external. The last column is the change in matrix rank that a deformed differential on the same generators would require.

A genuine deformation is a different object. An immersed Floer theory has two ordered branch-jump generators at each transverse self-intersection, of complementary degrees in a graded setup [AJ10]; a bounding cochain must satisfy the full Maurer–Cartan series, with repeated inputs at every order; and square-zero of the deformed differential is necessary, though not sufficient, for a proposed finite matrix. The search below labels the underlying double point instead. For a finite set S of self-intersections, a *support*, let $D_q^{\text{tab}}(S)$ add to D_q^{big} the candidate triangle and quadrilateral terms through *distinct* crossings of S , and put

$$h_q^{\text{tab}}(S) := \dim_{\mathbb{F}_2} C_q - 2 \operatorname{rank}_{\mathbb{F}_2} D_q^{\text{tab}}(S). \quad (\text{A.4})$$

Repeated insertions and higher operations are absent, and the accompanying Maurer–Cartan test uses only the candidate monogon, self-bigon and distinct-input self-triangle terms.

Computation A.2 (Low-order candidate supports). (i) At $q = 5$, among the singleton and two-element subsets of A_5 , the unique smallest support with $h_5^{\text{tab}} = 7$ is $S = \{s_A, s_B\}$, with $(\gamma, \theta) \approx (0.03, 1.27)$ and $(3.06, 4.98)$. One candidate quadrilateral changes the (x_4, y_6) bigon entry from 1 to 0; the resulting matrix has the sole entry (x_1, y_0) and squares to zero.

(ii) At $q = 7$, among the tested singleton crossings passing the low-order test, one crossing near $(0.042, 5.405)$ has $h_7^{\text{tab}} = 9$. Its candidate triangle changes one bigon entry from 1 to 0; the resulting matrix has entries $(6, 4)$ and $(7, 5)$ and squares to zero.

(iii) At $q = 11$ with the parameters $(0.05, 0.16, 0.40)$, 62 singleton crossings pass the low-order test and give matrix rank three. Exactly 56 of the resulting matrices square to zero; the square of each of the other six has a single nonzero entry, namely $(9, 15)$, $(9, 16)$, $(0, 14)$, and $(9, 14)$ for the remaining three. At the parameters $(0.03, 0.10, 0.25)$ the same test returns 66 singletons of the target rank, of which 16 fail square-zero and 50 pass.

At $q = 5$ the matrix D_5^{big} has rank two, and the pair s_A, s_B of Figure 2 is the only tested support that lowers the rank to one, so $h_5^{\text{tab}} = 9 - 2 = 7$; minimality holds only within this

search. At $q = 7$ the rank three of D_7^{big} drops to two and $h_7^{\text{tab}} = 13 - 4 = 9$; at the parameters $(0.07, 0.16, 0.40)$ the crossing is near $(0.058, 5.413)$ and gives the same entries. This crossing is S_{69} , whose type-D deformation is the four-arrow switch $b_{S_{69}}$ rather than a support labelled only by its double point, as in Computation A.2. Resolving both local sectors at the selected $q = 5$ and $q = 7$ crossings and recounting candidate bigons reproduces the matrices $D_q^{\text{tab}}(S)$ for one sector choice; at $q = 7$ Proposition 3.7 replaces this experiment by an exact calculation. Computation A.2 proves neither existence nor uniqueness of a bounding cochain: the candidates are not ordered branch jumps, the full Maurer–Cartan series is never tested, and neither 56 nor 50 counts bounding cochains.

A.4. Methods. All statements of Sections 3 and 4 that rest on computation are finite computations in exact arithmetic over $\mathbb{F}_2$, with the following exceptions, which are floating-point computations: the geometry of the piecewise-linear curves of Section 3 (their self-intersections and their crossings with the arc system, from which the encodings and the enumeration of smoothings are read off), which uses numerical tolerances; the finite winding test of §A.3, which evaluates winding functions on sampled regions; and the traced curves described below.

Morphism homology. Morphism complexes in $\text{Tw}(\mathcal{B})$ are computed in full, without truncating word length, by the mapping-cone reduction of Lemma 3.5, whose torsion condition is checked for each pair. A second, independent implementation computes $H\text{Mor}(X, Y)$ from the structure of $\text{Mor}(X, Y)$ as a module over the polynomial ring $\mathbb{F}_2[H]$ of [KWZ19]; it reproduces the dimensions of Propositions 3.7 and 3.8, and the dimensions of Section 4 were computed with it.

Encodings. Curves given by their lifts to the planar cover are encoded as twisted complexes by recording their crossings with the lifts of the arc system and the faces between consecutive crossings, as in §3.3.2. The smoothed curves of Proposition 3.7 were encoded with smoothing disks of radii 0.003, 0.006 and 0.009, with the same result. This encoder was compared with the invariants of 216 rational tangles computed by Zibrowius’s program [Zib]: in every case $\widetilde{\text{BN}}(Q)$ is the straight lattice arc and $\widetilde{Kh}(Q)$ the figure-eight around it. The identification of the earring curve E_r with $\Phi(\widetilde{Kh}(Q_{-r}))$ in §4.1 was checked by comparing encodings at 49 slopes, and the two sides of (4.1) were compared at the 340 slopes used for $q = 5$ and the 342 slopes used for $q = 7$ in Corollary 4.6; they agree at every slope. The invariants $\widetilde{\text{BN}}(T_q)$ and $\widetilde{Kh}(T_q)$ for $q = 3, 5, \dots, 21$ were computed with the same program, and the ranks $\text{rank } \widetilde{Kh}(K_r(q); \mathbb{F}_2)$ with Khoca 1.5 [Lew26].

Traced curves. The perturbed curves of Smith for $q = 5, 7, 11, 13, 17, 19$ were traced numerically from the equations of [Smi24, Prop. 4.1], composed with the lines of slopes $1/3$ and $1/q$ through $(0, 0)$, for several perturbation parameters in $[0.005, 0.02]$ and several step sizes, and transported to negative perturbation parameter by the symmetry $\mathfrak{s}$ of Remark 3.2. For $q \leq 13$ all traces at a given q have strictly isomorphic encodings, isomorphic to N_q . These are floating-point computations and serve as corroboration only.

Searches. Strict isomorphism of loop-type complexes is decided by comparing the words of the components up to rotation and reversal. Edit distances are computed by an exact search over the ways of cutting the word of one complex into intervals that reassemble to the word of the other; for $q = 17, 19$ the search excludes edit distances at most 5 and was stopped at 6. Chains of algebraic switches are enumerated breadth first, up to strict isomorphism, from both ends, and the two sides are compared. Each search is exhaustive within the class it states, and no conclusion is drawn beyond that class.

A.5. Reproducibility. The programs that carry out the computations of this paper are included as ancillary files with its arXiv submission, and they are also available in the public repository

<https://github.com/bwuebben/pillowcase-bounding-cochain>.

In the ancillary files the programs for Section 3 and §A.3 are in the directory `pillowcase/`, and those for Section 4 and the rest of this appendix are in `code/`; in the repository these are the directories `pillowcase/` and `paper2b/code/`. Each program prints the numbers of the statements it checks next to the computed values and reports whether the two agree, and the file `README.md` in each directory lists the statements that each program checks; there, oriented and unoriented smoothings are called census and opposite smoothings, and connector–inherited self-intersections connector–main ones. The programs use only the Python standard library. The invariants computed with Zibrowius’s program [Zib] and the ranks computed with Khoca [Lew26] are included as data, together with their input files, so the checks run without these programs. The instanton dimensions used throughout are [Wue26, Thm. 1.1] and Proposition 3.8, and they are independent of all of this code.

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